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SLC4A1 mutations that cause distal renal tubular acidosis alter cytoplasmic pH and cellular autophagy

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eLife Assessment

This work reports the characterization of newly identified genetic variants of SLC4A1 in patients with distal renal tubular acidosis. Cell culture studies supplemented with histological analysis of a previously established disease mouse model provide **convincing** evidence that some of the variants increase intracellular pH, reduce ATP synthesis, and attenuate autophagic degradative flux. The study is **valuable** in establishing a mechanistic framework for future exploration of the link between intracellular pH and mutations in SLC4A1 in vivo.

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Abstract

Distal renal tubular acidosis (dRTA) is a disorder characterized by the inability of the collecting duct system to secrete acids during metabolic acidosis. The pathophysiology of dominant or recessive *SLC4A1* variant related dRTA has been linked with the mis trafficking defect of mutant kAE1 protein. However, *in vivo* studies in kAE1 R607H dRTA mice and humans have revealed a complex pathophysiology implicating a loss of kAE1-expressing intercalated cells and intracellular relocation of the H⁺-ATPase in the remaining type-A intercalated cells. These cells also displayed accumulation of ubiquitin and p62 autophagy markers. The highly active transport properties of collecting duct cells require the maintenance of cellular energy and homeostasis, a process dependent on intracellular pH. Therefore, we hypothesized that the expression of dRTA variants affect intracellular pH and autophagy pathways. In this study, we report the characterization of newly identified dRTA variants and provide evidence of abnormal autophagy and degradative pathways in mouse inner medullary collecting duct cells and kidneys from mice expressing kAE1 R607H dRTA mutant protein. We show that reduced transport activity of the kAE1 variants correlated with increased cytosolic pH, reduced ATP synthesis, attenuated downstream autophagic pathways pertaining to the fusion of autophagosomes and lysosomes and/or lysosomal degradative activity. Our study elucidated a close relationship between the expression of defective kAE1 proteins, reduced mitochondrial activity and decreased autophagy and protein degradative flux.

Introduction

Distal renal tubular acidosis (dRTA) is a disorder characterized by the inability of the collecting duct system to secrete acids during metabolic acidosis¹. In addition to hyperchloremic metabolic acidosis, patients with this disease can present with hypokalemia, kidney stones, urinary sodium waste and difficulty to thrive. Expression of pathogenic variants in the *ATP6V0A4*, *ATP6V1B1*, *FOXI1*, *WDR72* and *SLC4A1* genes are the usual genetic aetiologies^{2–4}. The *SLC4A1* gene encodes the anion exchanger 1 (AE1) protein, which is an electroneutral chloride/bicarbonate exchanger⁵. It exists in two forms: a 911 amino acid erythroid isoform known to interact with erythroid cytoskeletal proteins and participate in red cell respiration and integrity, and a 65 amino acid (NH₂-terminal) truncated isoform primarily found in the basolateral membrane of renal type A intercalated cells (A-IC)^{6,7} and podocytes⁸. This isoform participates in bicarbonate reabsorption and through its physical and functional interaction with the cytosolic carbonic anhydrase II and apical H⁺-ATPase, it supports apical proton export and urine acidification⁹.

Pathogenic variants in the *SLC4A1* gene can result in either red cell defects (such as Southeast Asian ovalocytosis^{10,11} and hereditary spherocytosis¹²), renal cell defects (dRTA¹⁰) or both in patients with homozygous (Band 3 Coimbra and Band 3 Courcouronnes^{13,14}) or compound heterozygous variants^{15,16}. Renal *SLC4A1* disease-causing variants have only been found in the transmembrane domain - where it could impact protein structure and its transport function- or in the short carboxyl domain - where it possibly affects protein-protein interactions. The pathophysiology of dominant or recessive *SLC4A1* related dRTA (hereafter named dRTA) has originally been linked with the mis-trafficking defect of mutant kAE1 protein^{10,17}. However, recent *in vivo* studies in R607H (orthologous to human R589H dRTA variant) and L919X knockin mice and dRTA patients have revealed a complex pathophysiology where kAE1-expressing intercalated cells were lost, and in the remaining type-A intercalated cells, the H⁺-ATPase relocated intracellularly and accumulated autophagy marker p62 and ubiquitin-positive material¹⁸.

The highly active transport properties of collecting duct cells require the tight maintenance of cellular energy and homeostasis. The autophagy-mediated turnover of damaged organelles is necessary for protecting collecting duct cells as in most renal cells¹⁹. The chloride/bicarbonate exchange function of kAE1 in A-ICs confers a pivotal role in pH homeostasis and thus is a major contributor to cellular homeostasis. kAE1 protein interacts with several proteins such as integrin-like kinase (ILK), adaptor-related protein complex 1, 3 & 4 (AP-1, AP-3 & AP-4 mu1A), transmembrane protein 139 (TMEM139), kinesin family member 3B (KIF3B), and clathrin among others^{7,20–22} that support protein stability and trafficking. It also interacts with homeostatic proteins including the glycolytic enzyme glyceraldehyde-3-phosphate dehydrogenase (GAPDH)²³ and the antioxidant enzyme peroxiredoxin 6 (PRDX 6)²⁴, which play major roles in cellular energy metabolism and oxidative stress response, respectively.

In this study, we report the characterization of newly identified dRTA genetic variations and provide evidence of abnormal autophagy and degradative pathways in cells and kidneys from mice expressing dRTA mutant kAE1 proteins.

Concise Methods

Ethics Approval

This study was conducted in accordance with all national and institutional animal care guidelines and approved by the University of Alberta's Animal Care and Use Committee (AUP #1277).

Newly identified *SLC4A1* variations from dRTA patients

The patient carrying the R295H mutation was a boy carrying the variation in the homozygous state, whose genetic diagnosis was made at the age of 5, following growth retardation of ~2 SD for both weight and height, with bicarbonate at 18 mmol/L, potassium at 2.8 mmol/L, chloridemia at 94 mmol/L, calcemia at 2.55 mmol/L, and a urinary pH of 7.5. He also had a history of a pyeloureteral junction syndrome that was surgically managed at the age of 3. The R295H dRTA

variation is a nonsense homozygous substitution characterized by a replacement of guanine (G) on position 884 by adenine (A) in the coding sequence. It has an allelic frequency of 0.14 % in the European population.

For the Y413H variant, the patient was a female diagnosed at 1 month old, with an urinary pH of 7.5, evidence of nephrocalcinosis and failure to thrive. The S525F variant has been previously reported²⁵, but in brief, the patient was a 13 year old female with plasma pH of 7.25, plasma bicarbonate at 15.3 mmol/L who also presented with polyurethral junction syndrome, nephrocalcinosis and nephrolithiasis since childhood. The Y413H and S525F dRTA variations are nonsense heterozygous substitutions characterized by a replacement of thymine (T) at position 1237 by cytosine (C) and a replacement of cytosine (C) at position 1574 by thymine (T), respectively. The R589H dRTA variation has been previously described¹⁸. No follow up data were available for all patients.

Mice

Transgenic mice carrying the R607H knockin (KI) mutation (murine equivalent to human R589H mutation) were previously described¹⁸. Homozygous mice used throughout the study display incomplete dRTA with alkaline urine without metabolic acidosis at baseline as previously reported¹⁸. Homozygous mice or wild-type (WT) littermates were fed a standard rodent chow (Picolab® Rodent Diet 20 # 5053, LabDiet, ST. Louis, MO, USA) or for Figure 5 [H-K](#), a salt-depleted diet with acid challenge as previously reported²⁶ with adequate and constant water supply, and maintained on a 12-hour light and dark cycle throughout their lifespan.

Statistical analysis

All the experiments were independently repeated a minimum of three times. Experimental results were analyzed using the GraphPad Prism software and are summarized as mean ± SEM. All statistical comparisons were made using unpaired Student's t test or one/two-way ANOVA followed by a post-hoc test as indicated in figure legends. A P-value less than 0.05 was considered statistically significant. All datasets were assessed for normality, and outliers identified by Prism were excluded.

Detailed Material and Methods are listed in the Supplementary Material.

Results

The dRTA kAE1 variants traffic to the basolateral membrane but have reduced transport activity in mIMCD3 cells

We first characterized three newly identified dRTA mutations and compared them with kAE1 WT or previously characterized kAE1 R589H mutant. [Figure 1A](#) depicts the alpha fold predicted structure of kAE1 showing amino acids mutated in each kAE1 mutant. kAE1 R295H is a recessively inherited substitution in the N-terminal cytosolic domain of the protein. kAE1 Y413H is a dominantly inherited substitution in transmembrane domain (TM) 1 of the core domain. In the gate domain, dominantly inherited S525F and R589H substitutions occur in TM 5 and TM 6, respectively. We generated the newly identified dRTA variant cDNAs and expressed them or kAE1 WT in mIMCD3 cells. As seen on [Figure 1B](#), kAE1 R295H, S525F and R589H variants display 2 typical bands similar to kAE1 WT. The top band (open circle) encompasses kAE1 proteins carrying a complex oligosaccharide that have reached the Golgi and beyond, while the bottom band (black circle) corresponds to high mannose-carrying kAE1 proteins located in the endoplasmic reticulum. However, kAE1 Y413H mutant bands intensity was overall weaker than WT and displayed a predominant single band aligned with high mannose-carrying kAE1 proteins. These results indicate that the 3 newly described dRTA mutants are successfully expressed in mIMCD3 cells. We next localized these mutants by immunofluorescence in polarized mIMCD3 cells. Both kAE1 WT and mutants appropriately co-localized with basolateral membrane marker beta-catenin in polarized mIMCD3 cells ([Figure 1C](#)). kAE1 R589H location has previously been reported at the

basolateral membrane in polarized mIMCD3 cells¹⁸. However, staining for kAE1 Y413H was again weaker and seemed more intracellular than other mutants. To address a possible premature degradation of this variant, we measured its lifetime and observed that its degradation begins 6 hours post-synthesis while kAE1 WT abundance remained stable for 24 hours (Figure 1D). Finally, to assess the transport activity of the mutants, we examined the steady-state cytosolic pH (pHi) and rate of intracellular alkalinisation of mIMCD3 cells expressing kAE1 WT or mutants (Figure 1E-G). Using BCECF-AM, we observed that the steady state intracellular pH (pHi) of kAE1 mutant cells was more alkaline than WT cells (except for kAE1 R295H cell pHi, which has a similar trend but is not significantly different from WT) (Figure 1F), in agreement with rate of intracellular alkalinisation (Figure 1G). Note that transport activity, measured as rate of alkalinisation, is measured by reverting kAE1 exchange activity from bicarbonate *export* to *import* (see Methods, protocol for transport assay), hence a reduced alkalinisation rate is observed in cells expressing defective kAE1 protein compared to WT (panel G). Overall, these results indicate that except for kAE1 Y413H mutant, the other newly described variants are expressed and traffic to the basolateral membrane of polarized mIMCD3 cells, similar to the previously published kAE1 R589H mutant. Given that the premature degradation of kAE1 Y413H mutant likely explains the dRTA phenotype, we did not perform further assays on cells expressing this protein.

Autophagy processes are altered in mIMC3 cells expressing the kAE1 R295H, S525F and R589H dRTA mutants and in R607H knockin kidney lysates

In mice expressing kAE1 R607H (the murine equivalent to human dRTA R589H substitution), a striking reduction in type-A intercalated cells was noted and in the remaining cells, autophagy marker p62 and ubiquitin accumulated in these abnormally enlarged cells¹⁸. We therefore investigated the autophagy machinery in dRTA mutant mIMCD3 cells and in R607H knockin (KI) mice. We first examined the ratio of autophagosome marker LC3BII protein relative to the total intensities of LC3BI and LC3BII as well as p62 levels. These experiments were performed at steady state, upon autophagy induction by starvation, or after autophagy inhibition by Bafilomycin (Baf) A1 (Figure 2A-C). Figure 2A – I shows a consistent increase in the ratio of LC3B II to total LC3B (I + II) in the mutant cells at steady state (panel D), with starvation (panel F) and with Baf A1 (panel H) except for the kAE1 R295H mutant which was not significantly different from WT. This suggests an altered autophagy process in the mutant cells, in agreement with preliminary findings from Mumtaz and colleagues¹⁸. There was no significant difference in p62 abundance in mutant cells compared to WT at steady state and with starvation (Figure 2E&G). However, with Baf A1, R589H mutant cells had significantly lower p62 abundance compared to WT (Figure 2I). To confirm these findings, we next assessed the abundance of these markers in whole kidney lysates from the kAE1 R607H KI mice. We observed that the total LC3B abundance was significantly higher in homozygous KI mice (KI/KI) compared to WT, with no difference in p62 (Figure 2J-L). Overall, these results suggest an abnormal autophagy in mutant mIMCD3 cells and in kidneys of R607H KI mice. Given the lack of difference in phenotype between the recessive kAE1 R295H and kAE1 WT mIMCD3 cells, we focused the subsequent experiments on dominant kAE1 S525F and R589H mutant cells. Further investigations will be needed to understand the pathophysiology associated with the kAE1 R295H novel variant.

Late autophagy steps are blocked in dRTA kAE1 mutant-expressing cells due to their alkaline intracellular pH

Given these preliminary findings of abnormal autophagy in dRTA mutant cells, we examined in more detail their autophagy machinery by transiently transfecting them with the eGFP-RFP-LC3 construct and monitoring the rate of autophagosome and autolysosome formation. In this assay, the green fluorescent protein (eGFP) fused to LC3 is quenched in the acidic environment of the autolysosome, while both eGFP and red fluorescent protein (RFP) fluoresce in vesicles in the neutral lumen of the autophagosome^{27,28}. To avoid the poor efficiency of transient transfection in

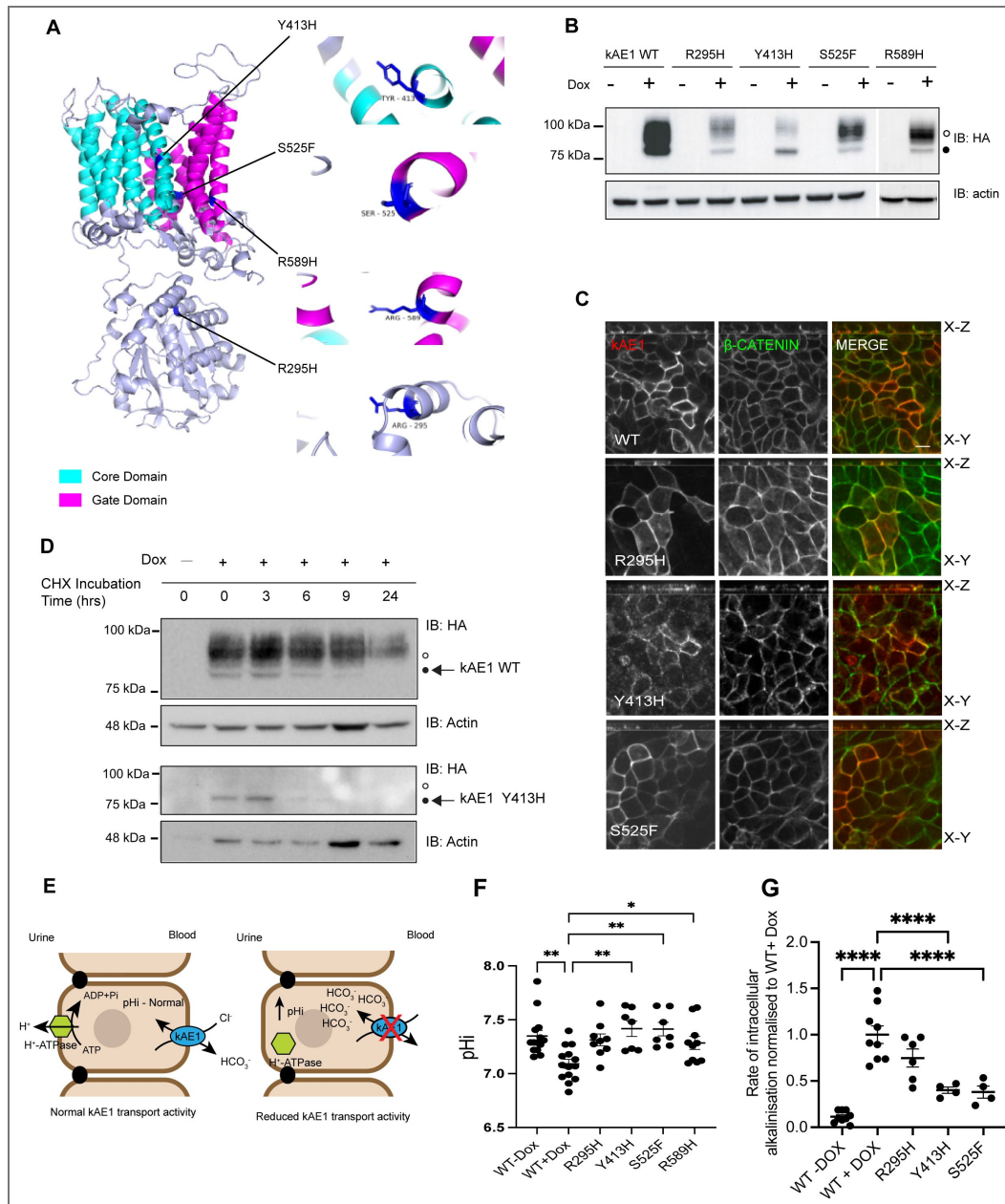


Figure 1. kAE1 R295H, Y413H, S525F and R589H dRTA mutants are either dysfunctional or prematurely degraded.

(A) Alpha Fold predicted structure of the kidney isoform of Band 3 anion exchanger 1 (kAE1) with core and gate domains highlighted. The dRTA kAE1 mutation sites are coloured in blue with line extensions detailing specific amino acids mutated. (B) Immunoblot showing expression of kAE1 WT, R295H, Y413H, S525F and R589H and corresponding actin band in mIMCD3 cells treated with and without doxycycline for 24 hrs. Mouse anti-HA antibody was used to detect kAE1-HA, top (open circle) and bottom bands (closed circle) correspond to kAE1 carrying complex and high mannose oligosaccharides, respectively. (C) Immunostaining of kAE1 WT or mutant (red) and β -catenin (green) in polarized mIMCD3 cells. Scale bar = 10 μ m. Red = kAE1, Green = β -Catenin. (D) Immunoblot of cycloheximide (CHX) chase assay with corresponding actin in kAE1 mIMCD3 WT and Y413H cells showing the degradation of kAE1 Y413H after 3hrs CHX incubation. (E) Cartoon depicting the transporter activity and expected changes in pHi in cells expressing kAE1 WT (left) or inactive mutant (right). (F) Graphical representation of intracellular pH (pHi) measurement of mIMCD3 kAE1 WT, R295H, Y413H, S525F and R589H cells. Error bars correspond to mean $\pm$ SEM, n = minimum 32. * P <0.05, ** P <0.01 using one-way ANOVA followed by a Dunnett's post-hoc test. (G) Rate of intracellular alkalinisation in WT or mutant mIMCD3 cells normalized to WT+ Dox. **** indicates P < 0.0001 using one-way ANOVA followed by a Dunnett's post-hoc test. Error bars correspond to mean $\pm$ SEM, n= minimum 4

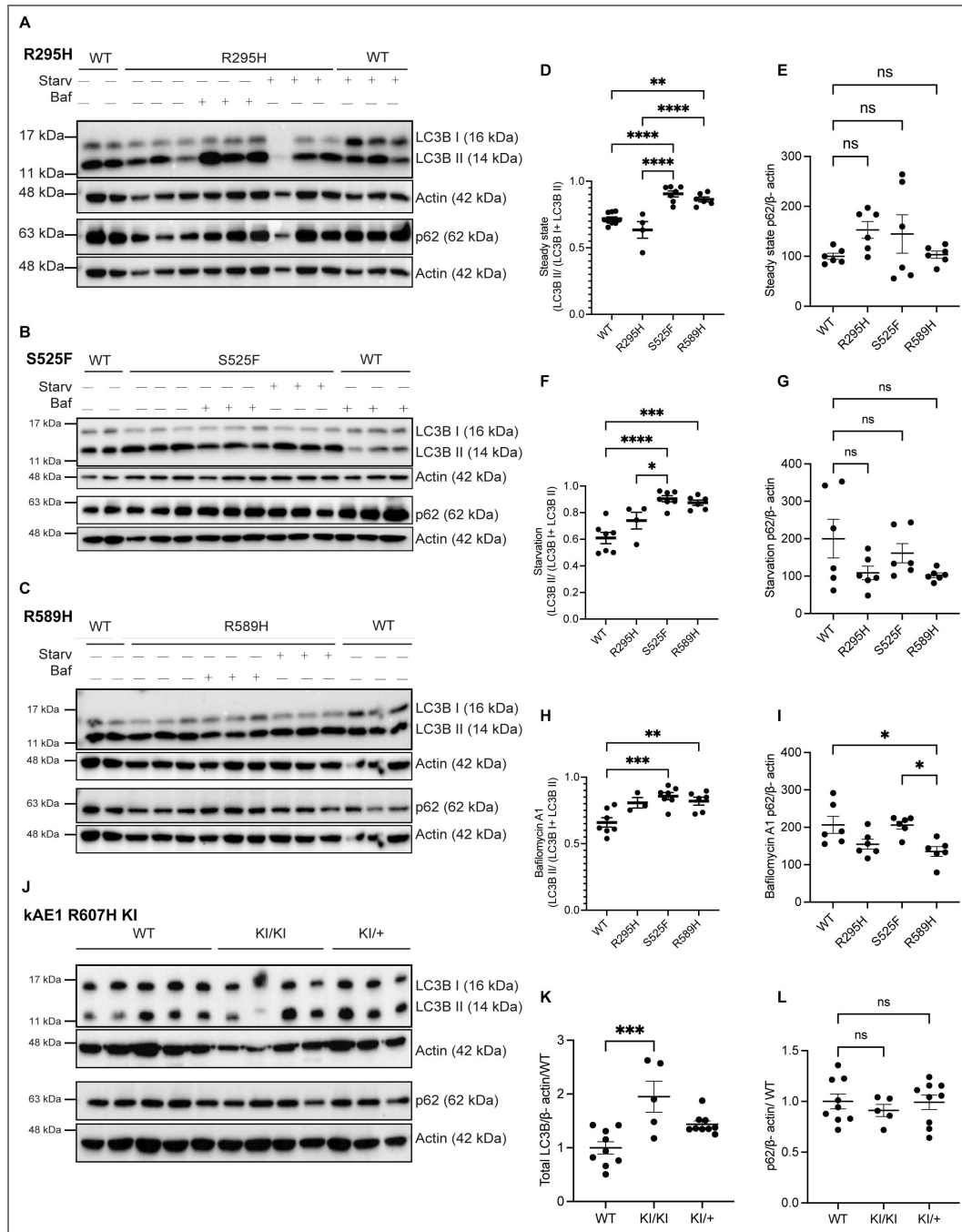


Figure 2. Autophagy is upregulated in dRTA kAE1 mutants *in vitro* and *in vivo*.

(A-C) Representative immunoblots of LC3B and p62 with corresponding actin abundance in kAE1 WT, R295H, S525F and R589H mIMCD3 cells at steady state, under starvation (Starv) or 400 nM Bafilomycin A1 (Baf) treatment. Note that p62 and LC3B were detected on the same blot for panel A and C, therefore the same actin blot is shown for both panels. (D-I) Quantification of all immunoblots showing the ratio of LC3B II to total LC3B and p62. Error bars correspond to mean $\pm$ SEM, $n = 3-8$. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.005$, **** $P < 0.001$ using one-way ANOVA followed by a Tukey's post-hoc test. Immunoblots (J) and quantification (K, L) of LC3B and p62 abundance in kAE1 R607H KI mouse whole kidney lysates. Error bars correspond to mean $\pm$ SEM, $n =$ minimum 5. **** $P < 0.005$ using one-way ANOVA followed by a Tukey's post-hoc test.

polarized cells, experiments were conducted in sub-confluent cells pending that the mutant proteins are present at the plasma membrane. We therefore performed cell surface biotinylations on 70-80 % sub-confluent cells, which confirmed a robust plasma membrane abundance of both kAE1 R589H and S525F that was not significantly different from the WT protein (Figure 3A&B). Therefore, we next assessed the efficiency of the autophagy machinery in WT or dRTA mutant-expressing cells (Figure 3C-F). Focusing on cells expressing either kAE1 WT, kAE1 S525F or R589H, we quantified the number RFP+ (red, acidic autolysosomes) and double eGFP+/RFP+ (yellow, not acidic autophagosomes) vesicles. kAE1 S525F mutant cells have significantly more autophagosomes than WT counterparts (Figure 3D), and both kAE1 S525F and R589H have significantly more autolysosomes than WT (Figure 3E). Figure 3F shows that both mutants had significantly more autophagosomes (yellow) than autolysosomes (red). This finding suggests an upregulation of autophagy and inhibition in the downstream steps of the process that involves the fusion of autophagosomes with the lysosome²⁹. As autolysosomes require luminal v-H⁺-ATPase-dependent acidification to efficiently clear cell debris³⁰, the higher intracellular pH seen in mutant cells (Figure 1F) may be detrimental to v-H⁺-ATPase full activity and impair proper autolysosomal acidification³¹. We therefore wondered whether chemically acidifying the pHi in mutant cells would rescue the autophagy machinery (Figure 3G-K). We first determined that incubation of mIMCD3 cells in 0.033 μ M nigericin in cell culture medium at pH 6.6 for 2 hours acidified cytosolic pH to 6.9 without causing cell death (Supplementary Figure 1A-C). Next, we observed that chemically reducing pHi to 6.9 in mutant expressing cells reduced the ratio of LC3B II to total LC3B (Figure 3J) and the abundance of lysosomal-associated membrane protein 1 (LAMP1) in R589H cells (Figure 3K) to levels similar to WT cells at steady state. These findings suggest that abnormal autophagy in the mutant cells may be caused by their alkaline pHi, resulting from a reduced anion exchange activity of the mutant kAE1 protein.

mIMCD3 cells expressing dRTA kAE1 mutants and R607H KI kidney tissues have abnormal lysosome number and size

The accumulation of autophagosomes and autolysosomes as seen above may suggest one or a combination of the following: an inability of autophagosomes to fuse with lysosomes and/ or a defect in lysosomal degradative activity in the mutant cells^{19,32}. We first examined the lysosomal degradative activity by assessing lysosomal protease Cathepsin B activity using Magic Red staining, a probe that fluoresces upon lysosomal protease cleavage (Figure 4A-B). In agreement with increased RFP+ vesicles (Figure 3E), the kAE1 S525F mutant cells had a significantly higher number of Magic Red positive vesicles than WT whereas the kAE1 R589H mutant cells had significantly larger Magic Red positive vesicles, suggesting an accumulation of undigested material³³. To validate this finding *in vivo*, we performed immunostaining and quantified LAMP1 positive staining in β 1 ATPase positive cells (a marker of ICs) in WT and R607H KI mouse kidney sections. The KI mice showed significantly more and larger LAMP1 positive vesicles compared to WT mice in both cortex and medulla (Figure 4 C-F). We probed further into the lysosomal activity by quantifying lysosomal protease Cathepsin D (immature, intermediate and mature) protein abundance by immunoblot in isolated primary murine A-ICs. Although the abundance of immature and intermediate cathepsin D did not differ between genotypes, the KI mice showed a significantly decreased abundance of mature cathepsin D (Supplementary Figure 2). Thus, in line with *in vitro* findings, A-IC from homozygous R607H KI mice display relatively more and larger lysosomes with reduced active protease abundance than WT littermates, suggesting a lysosomal defect in the dRTA kAE1 mutant cells.

dRTA kAE1 mutant cells have lower ATP production rate and abnormal mitochondrial content

Lysosomal degradation is highly dependent on a low luminal pH generated in part by the vacuolar-type H⁺-ATPase³⁴ whose activity depends on ATP hydrolysis. We therefore analysed the ATP production rate in mIMCD3 cells, specifically glycolysis and oxidative phosphorylation. We measured the oxygen consumption rate (OCR) (Figure 5A) and extracellular acidification rate

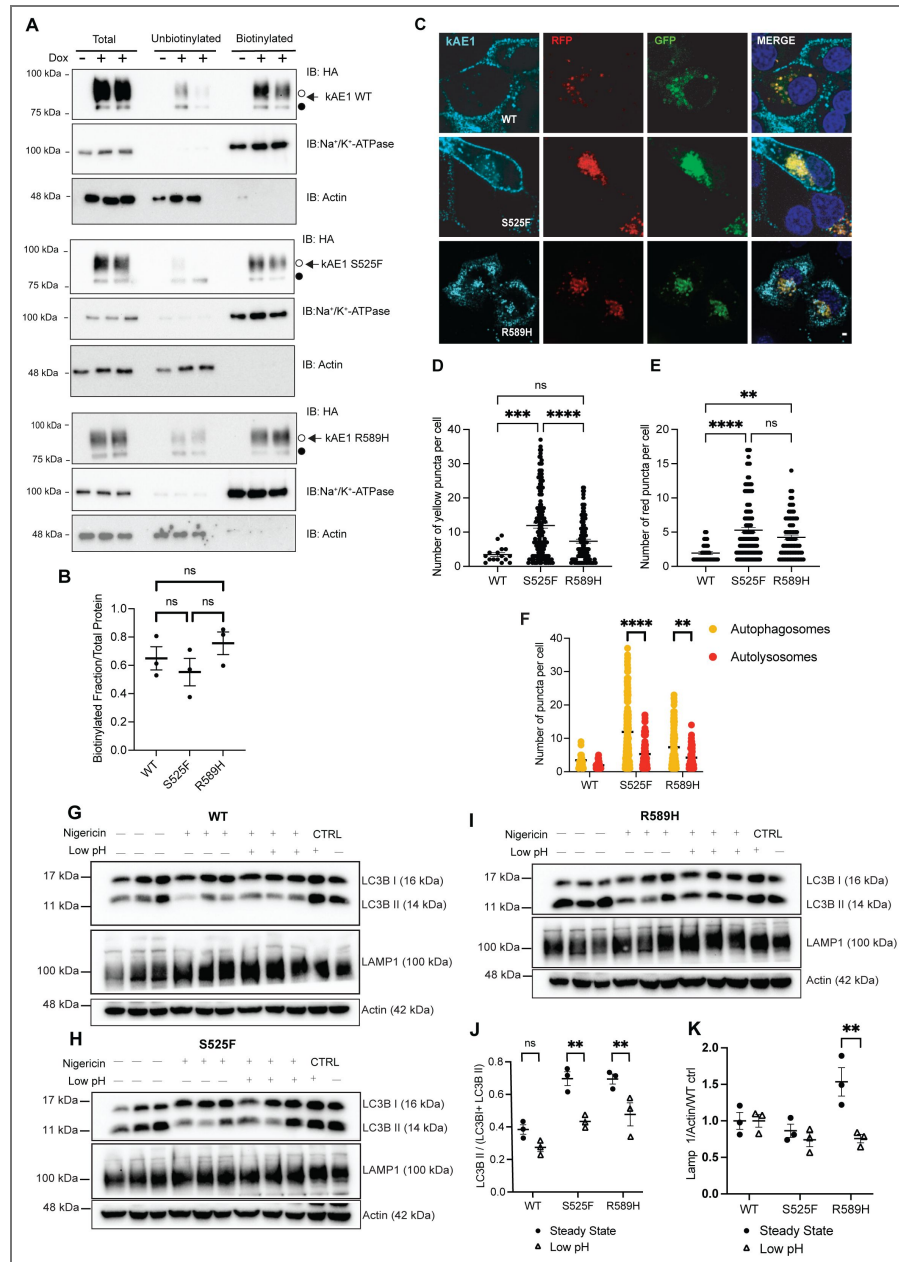


Figure 3. dRTA KAE1 mutants have more alkaline steady-state intracellular pH and altered autophagy flux.

(A) Representative immunoblots of cell surface biotinylation experiments in mIMCD3 cells expressing kAE1 WT or mutants (top panels), with control staining of plasma membrane marker Na⁺/K⁺-ATPase (middle panel) and intracellular marker actin (bottom panel). (B) Quantification of 3 independent cell surface biotinylation experiments. Data are represented by a single representative blot for each variant. N.s., not significant using one-way ANOVA. Error bars correspond to mean ± SEM, n=3. (C) Immunofluorescence staining in eGFP-RFP-LC3 transfected mIMCD3 cells expressing kAE1. GFP = green, RFP = red, kAE1 = cyan, nuclei = dark blue (merge only). Scale bar = 2 μm. Graphical representation of number of yellow (autophagosomes) (D) and red (autolysosomes) (E) puncta per cell expressing kAE1. Error bars correspond to mean ± SEM, n= minimum 32. ** P<0.01, *** P<0.005, **** P<0.001 using one-way ANOVA followed by a Tukey's post-hoc test. (F) Grouped graph of the number of yellow (autophagosomes) and red (autolysosomes) puncta per cell expressing kAE1, respectively. Note that the statistical analysis displayed only compared yellow and red groups for simplification. Error bars correspond to mean ± SEM, n= minimum 32. ** P<0.01, **** P<0.001 using two-way ANOVA followed by a Sidak's post-hoc test. (G-I) Immunoblot of LC3B, LAMP1 and actin in kAE1 WT, S525F and R589H mIMCD3 cells at steady state and under chemically reduced intracellular pH conditions. Graphical representation of the ratio of LC3B II to total LC3B ratio (J) or LAMP 1 (K) at steady state versus at low pH in mIMCD3 kAE1 WT, S525F and R589H. Black circles indicate steady state cells and triangles indicate low pH cells. Error bars correspond to mean ± SEM, n= 3. ** indicates P<0.01 using two-way ANOVA followed by a Sidak's post-hoc test.

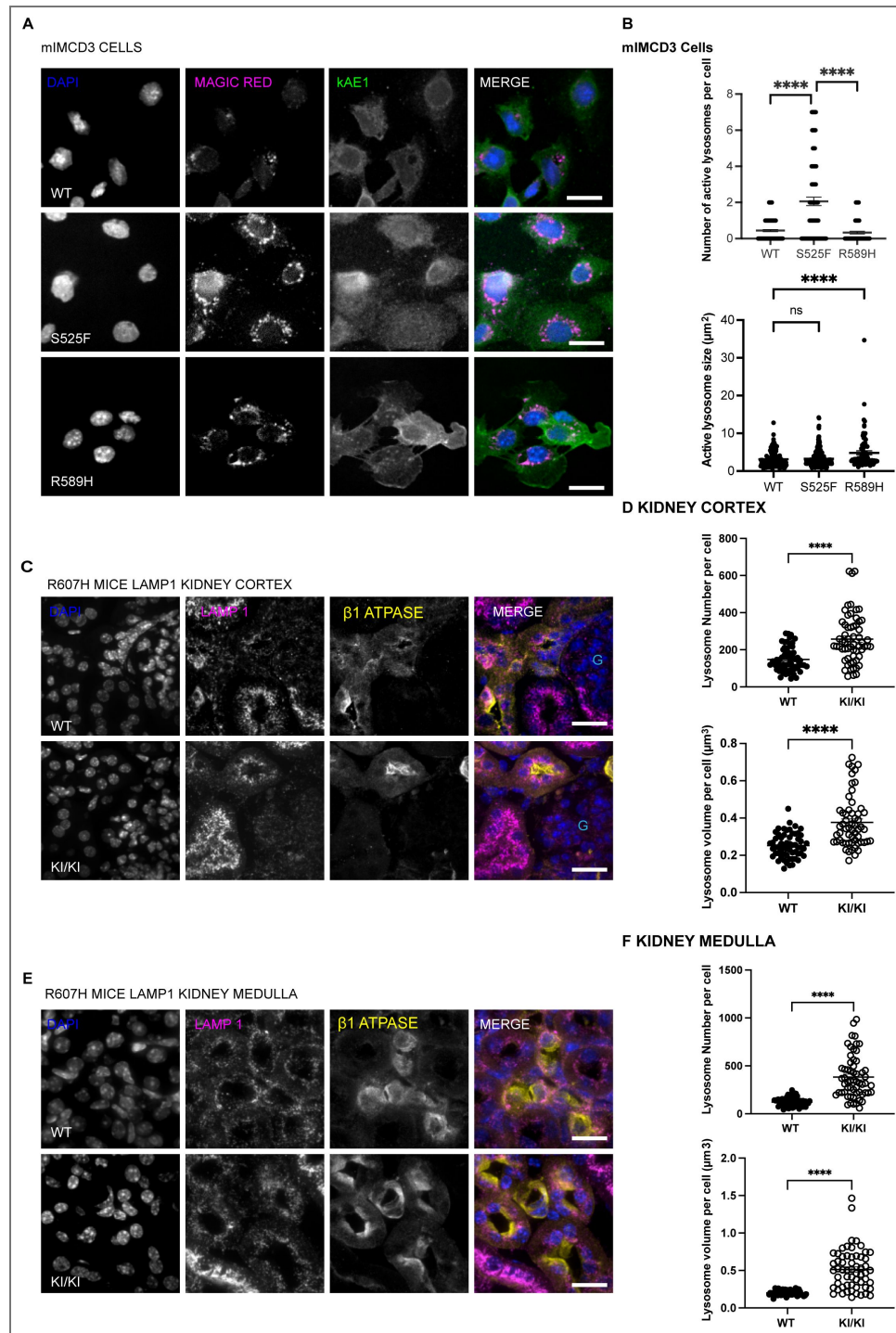


Figure 4. dRTA kAE1 mutants have bigger or more lysosomes than WT *in vitro* and *in vivo*.

(A) Immunofluorescence images of kAE1 WT, S525F and R589H mIMCD3 cells incubated with Magic Red substrate for 1hr at 37°C in the dark. Green = kAE1, magenta = active lysosomes, blue = nuclei. Scale bar = 16 μm . (B) Graphical representation of number and size of active lysosomes per cell. Error bars correspond to mean $\pm$ SEM, n= minimum 30. **** P<0.0001 using one-way ANOVA followed by Tukey's post-hoc test. Immunofluorescence images of LAMP1 and $\beta 1$ ATPase in kidney cortex (C) or medulla (E) from kAE1 WT and R607H KI mice. Blue = nuclei, magenta = LAMP1 (lysosomes), Yellow = $\beta 1$ ATPase, light blue "G" indicates the location of a glomerulus. Scale bar = 8 μm . Graphical representation of the number and volume of LAMP1 vesicles in $\beta 1$ ATPase positive cells in the kidney cortex (D) or medulla (F) of WT or R607H KI mice. Error bars correspond to mean $\pm$ SEM, n= 60. **** P<0.001 using Student's t-test.

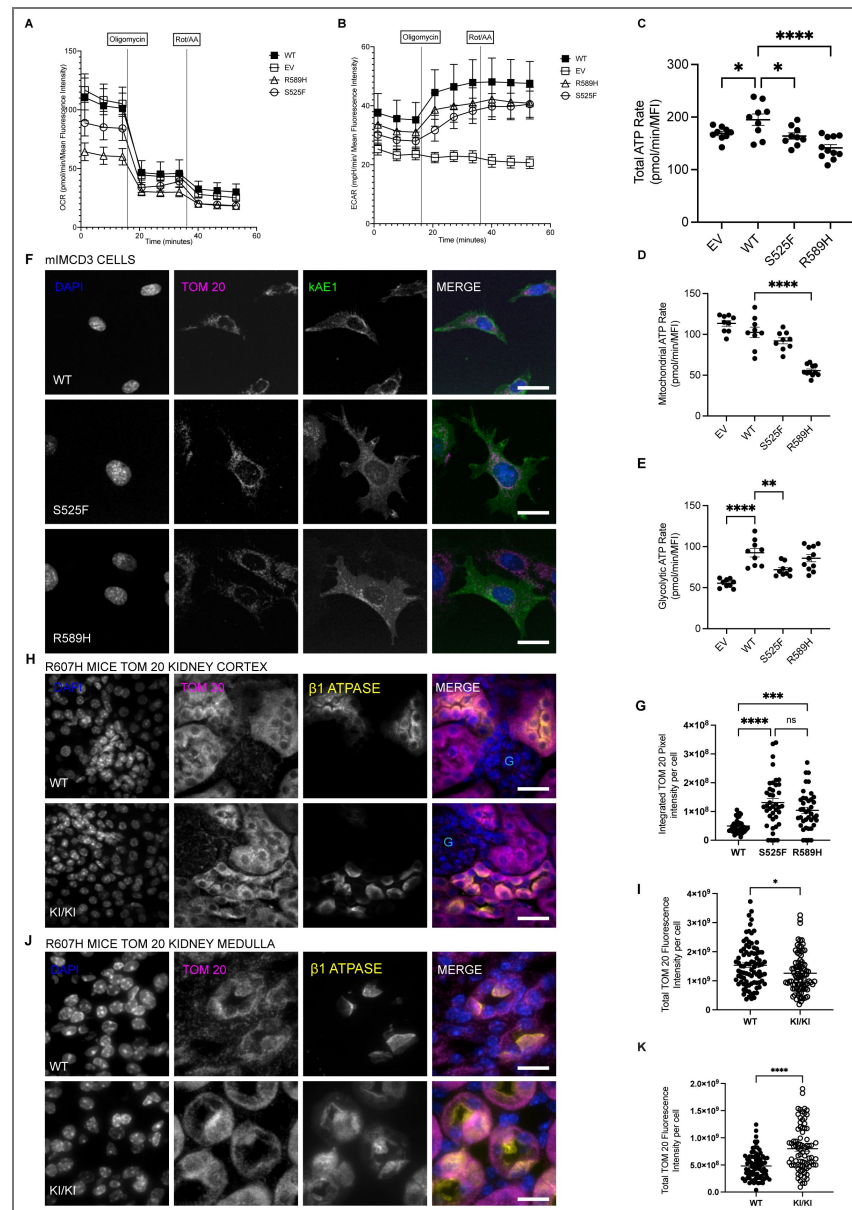


Figure 5. dRTA kAE1 mutant cells have lower ATP production rate and abnormal mitochondrial content.

Oxygen consumption rate (OCR) (A) and Extra Cellular Acidification Rate (ECAR) (B) of empty vector (EV) transfected cells, kAE1 WT, S525F or R589H mIMCD3 cells analyzed in a Seahorse XFe96 Extracellular Flux Analyzer with the ATP Rate Assay Kit. All cell lines, including EV transfected cells, were incubated with doxycycline, to eliminate a potential effect of doxycycline on measurements. (C) Graphical representation of the combination of ATP production rates from mitochondrial respiration (mitoATP) and glycolysis (glycoATP) of kAE1 WT, S525F and R589H mIMCD3 cells measured in real-time following sequential injections of oligomycin and Rotenone +Antimycin A. Error bars correspond to mean $\pm$ SEM, n= minimum. * $p < 0.05$, **** $p < 0.0001$ using one-way ANOVA followed by Tukey's post-hoc test. Graphical representations of mitochondrial respiration (D) and glycolytic ATP production (E) in kAE1 WT, S525F and R589H mIMCD3 cells. Error bars correspond to mean $\pm$ SEM, n= minimum 8. *** $p < 0.01$, **** $p < 0.0001$ using one-way ANOVA followed by Tukey's post-hoc test. (F) Immunofluorescence staining of TOM 20 and kAE1 in kAE1 WT, S525F and R589H mIMCD3 cells. Blue = nuclei, Magenta = TOM 20, Green = kAE1. Scale bar = 8 μ m. (G) Graphical representation of total TOM 20 fluorescence intensity per cell expressing kAE1. Error bars correspond to mean $\pm$ SEM, n= minimum 40. *** $p < 0.001$, **** $p < 0.0001$ using one-way ANOVA followed by Tukey's post-hoc test. Immunofluorescence images of TOM 20 and β 1 ATPase in kidney cortex (H) or medulla (J) of kAE1 R607H WT and KI mice exposed to a salt depleted diet with an acid challenge²⁶. Blue = nuclei, magenta = TOM 20 (mitochondria), Yellow = β 1 ATPase, light blue "G" indicates the location of a glomerulus. Scale bar = 8 μ m. Graphical representation of the total TOM 20 fluorescence intensity in β 1 ATPase positive cells in the cortex (I) or medulla (K) of the kidney. Error bars correspond to mean $\pm$ SEM, n= 90. * $p < 0.05$, **** $p < 0.0001$ using Student's t-test.

(ECAR) (Figure 5B) in empty vector transfected, kAE1 WT or mutant cells. Both kAE1 S525F and R589H mutant cells had a lower ATP production rate compared to WT (Figure 5C). More specifically, the R589H mutant cells had a lower mitochondrial ATP production rate (Figure 5D) whereas the kAE1 S525F mutant cells exhibited a lower glycolytic ATP production rate (Figure 5E). With the mitochondria being the major ATP-producing organelles³⁵, we assessed mitochondrial content by immunostaining of translocase of the outer membrane 20 (TOM20) both *in vitro* and *in vivo*. Both kAE1 S525F and R589H mutant cells have higher mitochondrial content compared to WT as determined by total overall intensity of TOM20 positive puncta (Figure 5F&G). In line with this result, although a decreased fluorescence intensity was observed in the cortex, there was a significantly higher TOM20 fluorescence intensity in medullary kidneys of homozygous R607H KI mutant mice compared to WT littermates (Figure 5H-K).

Discussion

In this study, we characterized three newly identified dRTA-causing kAE1 variations. Combining *in vivo* and *in vitro* studies, we demonstrated that reduced transport activity of the kAE1 mutants correlated with increased cytosolic pH, reduced ATP synthesis, attenuated downstream autophagic pathways and lysosomal dysfunction, pertaining to the fusion of autophagosomes and lysosomes and/or lysosomal degradative activity (See summary in Supplementary Tables 1 and 2).

In line with previous observations in mIMCD3 cells¹⁸, the kAE1 R295H, Y413H and S525F mutants were properly localized to the basolateral membrane after polarization. When expressed in Madin-Darby canine kidney (MDCK) cells, other dRTA-causing kAE1 mutants such as dRTA R602H, G701D, V488M, deltaV850 variants exhibited a plasma membrane trafficking defect^{10,36,37}. In contrast, the kAE1 R589H mutant was correctly targeted to the plasma membrane¹⁸. Functionally, cells expressing the Y413H and S525F mutants exhibit about 60 % reduction in chloride/bicarbonate exchange activity compared to kAE1 WT, similar to the previously characterized recessive G701D mutant but unlike the R295H mutant³⁸. Therefore, the mechanism causing dRTA remains unclear in the case of the newly identified recessive R295H variant. Additionally, the kAE1 Y413H variant exhibited a shorter half-life than WT counterpart, likely explaining dRTA, and thus was not further investigated. These findings add to the growing list of *SLC4A1* gene variations causing dRTA.

We next investigated the roots of the altered autophagy briefly reported in R607H KI mice¹⁸ using mIMCD3 cells and whole kidney lysates. In kidneys of the KI mice, a decrease in the number of A-IC, accumulation of p62 and ubiquitinated proteins and enlarged remaining A-ICs, suggested altered autophagy processes in dRTA mutant cells and in homozygous R607H KI mice^{18,38}. During the autophagy process, LC3B I (a marker for autophagosomes) is converted to lipidated LC3B II³⁹ and p62 aggregates to facilitate the degradation of ubiquitinated proteins within the autophagosome complex⁴⁰. In mIMCD3 cells expressing dRTA kAE1 S525F and R589H variants, LC3B lipidation was increased compared to WT, an increase that persisted with both Baf A1 treatment and starvation. Although opposite effects were expected under inhibition or induction of autophagy, such similar effect has been previously described. In the proximal tubule of obese mice, LC3B accumulation indicating a stagnated autophagy flux, was observed with both chloroquine treatment and 24 hour starvation⁴¹. Similarly, in NRK-52E cells, a disruption of the autophagy machinery in high cadmium-stressed cells resulted in LC3B II accumulation under either Baf A1 or rapamycin (an autophagy inducer) treatment⁴². Although LC3B II is elevated during both increased autophagy flux and disrupted autophagy, we did not observe significant accumulation of p62 in mIMCD3 cells expressing dRTA mutants. p62 is specifically a marker of autophagy-mediated protein clearance^{43–45}. Therefore, p62 accumulation in R607H KI mouse kidney sections strongly suggests a compromised autophagy-mediated clearance while the increased LC3B lipidation without significant changes in p62 in mIMCD3 cells points towards either an increased autophagic flux and/or a disrupted autophagy.

To obtain a clearer picture of the precise autophagic pathway altered in the dRTA mutants, we probed further into the different stages of autophagy and autophagy flux. We noted an accumulation of autophagosomes and autolysosomes in the S525F and R589H mutant cells. This

was recapitulated in the R607H KI mice which showed significantly more and larger LAMP 1 positive vesicles in both kidney cortex and medulla, suggesting a blockage in late steps of autophagy flux in both dRTA mutant cells and KI mice. Such blockage has been implicated in the pathophysiology of several diseases. In lysosomal storage disease, lysosome accumulation in proximal tubule cells is a key component in the pathways mediating epithelial dysfunction¹⁹. In this study, restoring autophagy flux attenuated disease progression. Another study in SK-N-SH, RT4-D6P2T and HeLa cells implicated autophagosome and lysosome accumulation in cellular toxicity associated with neurodegenerative diseases³². In agreement with altered lysosomal function, we also noted a greater abundance and size of active cathepsin B lysosomal protease vesicles in mIMCD3 cells. Increased cathepsin B activity affects lysosomal biogenesis, autophagy initiation and cellular homeostasis⁴⁶. In the renal context, cathepsin B knockout mice demonstrated a higher resistance and quicker recovery from glomerular damage⁴⁷, suggesting that cathepsin B accumulation may be detrimental to the cells. Increased cathepsin B abundance in the dRTA mutant cells also correlates with the accumulation of lysosomes. Overall, these findings suggest that the pathogenesis of dRTA in our *in vitro* and *in vivo* models involves an inhibition of autophagy flux at the downstream steps involving autophagosome-lysosome fusion and lysosomal protein clearance^{29,33,48,49}.

The question remained as to how these dRTA variants altered autophagy. The *SLC4A1-3* gene family that includes AE1 are regulators of intracellular pH in different cell types^{50–52}. The reduced anion exchange activity in mIMCD3 cells expressing the R589H and S525F variants expectedly correlated with an increased pHi compared to WT counterparts. Given that pHi variations can impact autophagy^{31,34,53–55}, we wondered whether this was the case for cells expressing kAE1 dRTA mutants. We observed that an acidic pHi in mutant cells restored autophagy levels similar to WT. Previous studies reported more perinuclear localization of lysosomes and autophagosome-lysosome fusion in cells with an increased pHi^{53,54}. Although not examined in our study, an increase in intracellular pH due to starvation decreased the levels of lysosomal kinesin superfamily member KIF2 and ADP-ribosylation factor-like 8B (ARL8), which are responsible for redistributing lysosomes to the cell periphery. This reduction subsequently inhibited mTORC1, resulting in increased autophagosome synthesis and autophagosome-lysosome fusion⁵³.

While an alkaline cytosolic pH partially explains the impairment in autophagy flux, a close relationship also exists between intracellular pH and cellular energy dynamics and metabolic stress, all of which are key regulators of autophagy⁵⁶. Therefore, it was plausible that kAE1 variant-induced autophagy dysregulation occurs through a signalling pathway akin to that of energy deprivation-induced autophagy. This hypothesis is supported by our findings of reduced ATP production rate in dRTA kAE1 S525F and R589H mutant cells compared to kAE1 WT cells. We also found that both kAE1 S525F and R589H mutant cells have higher mitochondrial content compared to kAE1 WT cells. The increased mitochondrial content coupled with low ATP point towards improper mitochondrial function in dRTA variant cells. Low ATP levels as seen in dRTA mutant cells may impair autophagy as was shown in human RPE cells⁵⁷. ATP reduction in RPE cells led to complex mitochondrial changes such as structural disorganization, enzyme activity decline, and oxidative damage to mitochondrial components and DNA⁵⁷. Similarly, in pancreatic islet cells, an alkaline pHi led to increased uptake of phosphate by mitochondria, accelerating the production of superoxide, promoting mitochondrial permeability transition, and inducing translational attenuation due to endoplasmic reticulum stress, ultimately impairing insulin secretion⁵⁸. Overall, our data support that expression of the kAE1 variants increases pHi which alters mitochondrial function and leads to reduced cellular energy levels that eventually attenuates energy-dependent autophagic pathways including autophagosome-lysosome fusion and lysosomal protein clearance.

In light of these observations, we postulated that correcting the alkaline pHi of dRTA mutant-expressing mIMCD3 cells will alleviate this blockage in autophagy flux. We observed that a chemically engineered pHi of 6.9⁵⁹ reduced LC3B II accumulation and LAMP1 abundance in mIMCD3 mutant cells to expression levels similar to that of WT cells at baseline. This suggests that a chemically reduced pHi facilitated protein clearance in the two dRTA mutant cells^{31,53} as noted

in other studies. In one such study, treatment of SH-SY5Y cells with FCCP and nigericin also acidified intracellular pH and triggered autophagy and mitophagy³¹. Similarly, acid loading in proximal tubular cells under chronic metabolic acidosis showed increased autophagic flux and mitophagy⁶⁰. These findings are in line with our results and establish a link between altered autophagy flux and the alkaline pHi of dRTA variant cells. Overall, our study provides one pathway (altered pHi) by which dRTA may arise. However, different variants induce different degree of functional defects as seen in Figure 1F & G. The kAE1 R295H, the only reported amino acid substitution in the amino-terminal cytosol causing dRTA, does not affect the transporter's function or pHi. Therefore, this variant may cause dRTA via a different pathway, for example defective protein-protein interactions, than transport-defective S525F or partially inactive R589H variants.

In conclusion, our study established a strong relationship between the expression of defective kAE1 proteins, reduced mitochondrial activity, decreased autophagy and impaired protein degradative flux. Whether this abnormal degradative pathway explains the premature loss of A-IC will need to be elucidated in further studies.

Supplementary Material

Material and Methods

Antibodies and chemicals

Mouse anti-HA antibody (hemagglutinin, Biolegend, formerly Covance), mouse anti- β -actin antibody (Sigma Aldrich or anti - β -actin HRP clone 2F1-1 Biolegend cat#643807), mouse anti-LC3B antibody (Cell Signaling), mouse anti-IVF 12s antibody (Developmental Studies Hybridoma Bank), rat anti-ATP6V1B1 antibody (BiCell #20901), rabbit anti-mTOR antibody and phospho mTOR antibody (Cell Signaling), Rabbit anti-4E-BP1 antibody and phospho 4E-BP1 (Ser65) antibody (Cell Signaling), mouse anti-p62 antibody (Abcam), mouse anti-p53 antibody (Cell Signalling), rabbit anti-Cleaved caspase 3 antibody (Cell Signalling), mouse monoclonal anti Na^+/K^+ -ATPase H-3 antibody (Santa Cruz Biotechnology, Dallas, TX), goat anti-mouse antibody horseradish peroxidase conjugated (HRP) (Cell Signaling Technology), Cy3-conjugated donkey anti-mouse antibody, anti-rabbit and anti-goat antibodies (Jackson Immunoresearch), X-tremeGENE HP DNA transfection reagent (Roche), Seahorse XFe consumables (Agilent).

Cell Lines, Transfections and Viral Infection

Mouse inner medullary collecting duct (mIMCD) cells (ATCC# CRL2123) were used for preparing kAE1 wild type and mutant cell lines. The pLVX TRE3G kAE1 construct was generated from the shuttling of human kAE1 cDNA with an external hemagglutinin (HA) epitope in position 557 (on eAE1) into pLVX-TRE3G plasmid (Clontech)⁶¹. This construct encodes a protein described as kAE1 throughout this paper. The kAE1 S525F and R589H mutants were generated with Q5 site-directed mutagenesis. All plasmids were introduced into the mIMCDs using a viral single shot packaging kit (Clontech).

Mouse intercalated cell isolation and tissue homogenate preparation

Kidney tissue homogenates were prepared as previously described⁶². Briefly, after decapsulation, freshly dissected kidneys were homogenized in cold lysis buffer (0.3 M Sucrose, 25 mM Imidazole, 1mM EDTA, 8.5 μ M Leupeptin, 1 mM PMSF), and vortexed over one hour every 15 min. Thie homogenates were then centrifuged at 14000 rpm for 15 minutes at 4 °C prior to measurement of protein concentration by Bicinchoninic Acid Protein Assay. Primary intercalated cells were prepared from homozygous kAE1 R607H transgenic mice. After cardiac perfusion with PBS, heparin, and collagenase B (Sigma Aldrich), kidneys were homogenized by MACS dissociation and intercalated cells enriched using CD 117 magnetic sorting (Miltenyi Biotec) as previously described⁶³. During the selection, intercalated cells were kept in MACS buffer (PBS, 2mM EDTA and and 0.5 % FBS). Cells were lysed with RIPA lysis buffer (2 mM EDTA, 2 % Deoxycholate, 0.3 M NaCl, 20 mM Tris/HCl pH 7.5, 2 % Triton X-100, 0.2 % SDS, pH 7.4), supplemented with complete

EDTA-free protease inhibitors, and PhoSTOP phosphatase inhibitor (Roche), PMSF, pepstatin, leupeptin and aprotinin. An aliquot was saved for bicinchoninic acid assay to determine protein concentration and remaining lysates were kept in Laemmli buffer at -20°C for immunoblotting.

Bicarbonate transport assay

This assay has been previously described⁶⁴. Briefly, confluent kAE1 WT-HA or mutant mIMCDs cells grown on coverslips were incubated with $1\text{ }\mu\text{g/ml}$ doxycycline (Sigma-Aldrich) for 18 - 24 hours at 37°C to induce kAE1 expression. They were then incubated with 2',7'-Bis-(2-Carboxyethyl)-5-(and-6)-Carboxyfluorescein, Acetoxymethyl Ester (BCECF – AM, Thermo Scientific), a fluorophore which excites at 440 and 490 nm and emits 510 nm wavelength for 30 minutes at 37°C . Using a fluorometer from Photon Technologies International (PTI) (London, Ontario, Canada), coverslips were perfused with NaCl-based Ringer's buffer (5 mM glucose, 5 mM potassium gluconate, 1 mM calcium gluconate, 1 mM magnesium sulfate, 10 mM HEPES, 2.5 mM sodium dihydrogen phosphate, 25 mM sodium bicarbonate, 140 mM sodium chloride) for 5 to 10 minutes. Once stable, initial fluorescence (corresponding to steady-state pH_i) was recorded for the first 30 seconds before switching to chloride-free containing sodium gluconate based Ringer's buffer of same osmolality. BCECF fluorescence was calibrated by perfusing cells with different pH buffers (6.5, 7, 7.5) in the presence of $10\text{ }\mu\text{M}$ nigericin. The Ringer's buffers were continuously bubbled with an air: CO₂ mixture (19:1), providing 5 % CO₂. Transport rates of the cells were determined by linear regression of the initial fluorescence variations (over the first 60 seconds), normalized to pH calibration measurements. All measurements were done using PTI FelixGX software.

Cell Treatments and immunoblotting

For autophagy experiments kAE1 WT or mutant cells were seeded to 70 % confluency on 10 cm culture plates and treated with $1\text{ }\mu\text{g/ml}$ doxycycline (Sigma-Aldrich) for 48 hours. Cells were then either treated with 400 nM bafilomycin A1 for four hours to inhibit autophagy or with Hanks balanced salt solution (HBSS, Gibco) to starve cells and induce autophagy or given no treatment. To chemically modify pH_i of cells, 90 – 100 % confluent cells were treated with $1\text{ }\mu\text{g/ml}$ of doxycycline (Sigma-Aldrich) overnight. Cells were then incubated in Ringers' buffer with pH 6.6, supplemented with $0.03\text{ }\mu\text{M}$ nigericin with a final potassium concentration of 168mM for two hours at 37°C . Steady-state cells were incubated in normal pH media without nigericin prior to lysis. Under treated conditions, pH_i was similar to pH_e (Supplementary Figure 1). Cells were lysed with RIPA lysis buffer (1% deoxycholate, 1mM EDTA, 0.15 M NaCl, 0.1% SDS, 10 mM TRIS/HCL (pH 7.5), 1% Triton X-100) with phosphatase inhibitors (cat. no. 04906837001; Roche PhosSTOP) and protease inhibitors (cat. no. 04693159001; Roche Complete Tablets, Mini EDTA-free) and stored at -20°C with or without 2x Laemmli buffer. The aliquot without Laemmli buffer was used for a bicinchoninic acid (BCA) assay to determine protein concentration. Following the BCA, 10 - 30 μg of total protein was loaded on SDS-PAGE gels. Proteins were transferred to PVDF membranes and antibodies listed above were used for detecting the proteins of interest. Primary antibodies were diluted in 1 % milk and incubated on membranes overnight at 4°C followed by secondary antibodies linked with horseradish peroxidase (HRP) for one hour at room temperature. Protein detection was done with the Enhanced Chemiluminescence reagent (ECL Prime, Invitrogen), and a BioRad Imager. The ImageLab software (BioRad) was utilized for the quantification of relative band intensities.

Cell Surface Biotinylation assay

mIMCD3 cells stably expressing kAE1 WT, S525F or R589H were seeded to 70-80% confluency. The cells were incubated with sulfo-N-hydroxysuccinimide-SS-biotin (Thermo cat.# 21331) (1.5 mg/ml in ice-cold PBS) for an hour at 4°C , quenched with 100mM glycine and lysed with RIPA lysis buffer, supplemented with complete EDTA-free protease inhibitors, and PhoSTOP phosphatase inhibitor (Roche), PMSF, pepstatin, leupeptin and aprotinin. Total protein levels were measured by BCA, and an aliquot was saved as "Total" fraction. 450 μg of each lysate was subsequently incubated with 100 μl streptavidin slurry beads for 1 h on a rocker at 4°C . Following centrifugation, the

supernatant was collected, and an aliquot kept as the “unbiotinylated” fraction. After six washes, the beads were resuspended in 50 μ l of 2X Laemmli buffer and incubated at room temperature for 30 min. The eluted biotinylated proteins were subsequently collected by centrifugation (“Biotinylated” fraction). The biotinylated fraction (45 μ l) was loaded on SDS-PAGE gel for immunoblot analysis along with 3 μ g of the Total fraction and a matched volume of unbound fraction per well. In addition to anti-HA antibody, the blots were probed for actin to ensure cell membrane integrity was intact during the biotinylation procedure, and for Na⁺/K⁺-ATPase as cell surface control.

Magic Red Assay

80 % confluent kAE1 WT or mutant mIMCD3 cells were treated with 1 μ g/ml doxycycline (Sigma-Aldrich) for 48 hours to induce kAE1 expression. Different treatments including: a 4-hour incubation with 400 nM bafA1 to inhibit autophagy, a 2-hour starvation in HBSS to induce autophagy or no treatment (steady state) were applied. Cells were then incubated with 1 % Magic Red reagent (ImmunoChemistry Technologies) in DMEM-F12 medium at 37°C in the dark for 30-60 minutes. Cells were then fixed with 4 % PFA, quenched with 100 mM glycine, permeabilized with 0.2 % Triton X-100, blocked with 1 % BSA and incubated with mouse anti-HA primary antibody and donkey anti-mouse Alexa fluor 488 conjugated secondary antibody for 30 mins each. Cells were then incubated with 4',6-diamidino-2-phenylindole (DAPI) for 5 mins before mounting using DAKO Mounting Medium (Agilent Technologies). A WaveFX confocal microscope was used to image the slides, and the images were analyzed blindly using the Fiji software.

Autophagy flux Assay

kAE1 WT-HA or mutant mIMCD3 cells seeded to 70 % confluency in a 6-well plate on coverslips were transiently transfected with the eGFP-RFP-LC3 cDNA construct (kind gift from Dr. Goping, Department of Biochemistry, University of Alberta) using the X-tremeGENE HP DNA transfection reagent (Roche). Four hours after transfection, the cells were incubated with 1 μ g/ml doxycycline for 48 hours at 37°C to induce kAE1 expression. Following this incubation, cells were incubated with blocking medium, mouse anti-HA primary antibody (1:200) and donkey anti-mouse Alexa fluor 649 (Jackson ImmunoResearch). Hoescht stain (ImmunoChemistry Technologies) was used to stain cellular nuclei. A WaveFX confocal microscope together with the Velocity and Fiji software were used to capture and analyze images.

Assessment of mitochondrial content

kAE1 WT-HA or mutant mIMCD3 cells seeded to 50 % confluency in a 6-well plate on coverslips were incubated with 1 μ g/ml doxycycline for 48 hours at 37°C and overnight in complete medium with no antibiotics. Cells were then fixed with 4 % PFA, quenched with 100 mM glycine, permeabilized with 0.2 % Triton X-100, blocked with 1 % BSA and incubated with mouse anti-HA and rabbit anti-TOM 20 primary antibodies and then with donkey anti-mouse Alexa fluor 488 and anti-rabbit Cy3 conjugated secondary antibodies. Cells were then incubated with 4',6-diamidino-2-phenylindole (DAPI) for 5 mins before mounting using DAKO Mounting Medium (Agilent Technologies). A WaveFX confocal microscope was used to image the slides, and the images were analyzed blindly using the Fiji software.

Metabolic flux analysis

kAE1 WT-HA or mutant mIMCD3 cells were treated with 1 μ g/ml doxycycline for 48 hours followed by an overnight incubation in complete media with no antibiotics and cells seeded at 2×10^4 cells per well in Seahorse XFe 96 well plates overnight to form a uniform monolayer. On the day of assay, culture medium was replaced with XF DMEM Medium pH 7.4 (103575-100, Agilent Technologies) with glucose (10 mM), pyruvate (1 mM) and L-Glutamine (2 mM) and incubated in a non-CO₂, 37°C incubator for one hour prior to their placement into the XFe96 Analyzer. Using the ATP production rate assay kit (#103592-100, Agilent Technologies) and XF cell Mito Stress Test kit (#103015-100, Agilent Technologies), metabolic indices were obtained from the Seahorse XFe96 Analyzer following manufacturer's procedures previously described²³. The total ATP rate is the sum of ATP production rate from both glycolysis and oxidative phosphorylation. Glycolysis

releases protons in a 1:1 ratio with ATP hence the glycolytic ATP rate is calculated from the glycolytic proton efflux rate (glycoPER). GlycoPER is determined by subtracting respiration linked proton efflux from total proton efflux by inhibiting complex I and III. The empty vector transfected cells provided a control for a potential effect of doxycycline on measurements. Oxygen consumption rate (OCR) and extracellular acidification rate (ECAR) were measured at various time points at basal state followed by injections of oligomycin (1.5 μM) and Rotenone +Antimycin A (0.5 μM).

Tissue Preparation and Immunostaining of kidney sections

Kidneys collected after perfusion were stored in 4 % PFA overnight at 4°C. The PFA solution was switched to 15 % sucrose for 2 hours and then transferred to 30 % sucrose solution overnight at 4°C. Thereafter, kidneys were fixed in O.C.T (Tissue-Tek®) and snap frozen in liquid nitrogen. These tissues were stored at -80°C until cryo-sectioning. Ten (10) micron tissue sections were fixed on a charged glass slide (ThermoFisher) and used immediately or stored at -80°C until immunostaining. For immunostaining, the slices were first air-dried for 20 mins, washed with PBS for 5 minutes and fixed with 4 % PFA for 20 mins at 4°C. The sections were quenched with 100 mM glycine for 15 minutes, permeabilized and blocked with 5 or 10 % serum in 0.2 % Triton in PBS for one hour at room temperature. Slices were incubated in primary antibody diluted in 5 or 10 % serum overnight at 4°C followed by secondary antibody diluted in 5 or 10 % serum for one hour at room temperature. Slices were washed with 0.1 % Tween 20 in PBS and incubated with DAPI for 5 mins at room temperature, mounted with DAKO mounting medium and sealed. Slides were air-dried and then stored at -20°C. Note that kidney sections analyzed in Figure 5H-K were obtained from mice fed a “salt-depletion with acid load” diet consisting of a low sodium and chloride diet for 8 days, complemented with 0.28 M NH_4Cl with 0.5 % sucrose in drinking water for 6 additional days as previously described². This diet triggered a metabolic acidosis, significantly lower plasma bicarbonate with a more alkaline urine in the homozygous KI mice compared to WT littermates.

Confocal imaging and image analysis

Immunofluorescent imaging was done with a WaveFX confocal microscope (Quorum Technologies, Guelph, Ontario, Canada) powered by a Volocity software (Quorum Technologies, Guelph, Ontario, Canada). Images were taken with 40 X oil immersion objective with z-stacks at 0.5- μm intervals. Quantitative image analysis was performed using the Volocity analysis software or by open-source cell image analysis softwares CellProfiler⁶⁵ and Fiji⁶⁶.

Image Analysis

Cell profiler was used to analyse TOM20 staining in mIMCD3 cells. After converting the three channel RGB images to grayscale using the split method, we manually outlined each cell in the channel corresponding to kAE1 staining and inputted back into the pipeline. The objects were then converted to a binary image and the TOM20 channel was used as the input channel to measure intensity of TOM20 staining in the manually outlined objects. The raw data including the sum of TOM 20 pixel intensity per cell were then analysed using GraphPad Prism software. A minimum of 30 cells were analysed.

Fiji software was used to analyse images from Magic Red staining and autophagy flux experiment. After each multichannel image was opened and merged in Fiji, the “multi point” tool was used to label all regions of interest. Freehand selection tool was used to manually draw the outline of all selected regions of interest, followed by the “Analyze” command to extract number and size of puncta in pixels. Pixel values were then converted to microns and data analysis was completed using GraphPad Prism. A minimum of 30 cells were analyzed.

Volocity software was used for analysis of kidney sections. From B1 H^+ -ATPase-positive single cells cropped from confocal images of medullary or cortical mouse kidney sections, the channel of interest was used to find objects using the “Find objects” command. Refinement within the selection was made based on object size. The minimum object size thresholds were 0.016 μm^2 for LAMP1 puncta and 0.02 μm^2 for TOM20 staining. This was then labelled as “population one”. From population one, touching objects were separated using a size guide of 0.02 μm^2 for LAMP 1 puncta

only with the “Separate touching objects” function. For TOM20 staining, all RFP positive stain within individual cells were characterized under one mask without separating touching objects. The minimum object size was set to $0.02 \mu\text{m}^2$ and everything less than that was considered background staining. The sum of TOM 20 fluorescence intensity per cell was collected and analysed using GraphPad Prism. For LAMP1 images, an exclusion criterion removing objects lesser than or equal to $0.02 \mu\text{m}^2$ was used to remove small background objects. Once the regions of interest in the image were properly outlined, data including number of puncta, intensity of puncta, area/ volume of puncta, among other measurements were exported and data analysed with GraphPad Prism. A minimum of 60 cells were analysed.

<i>dRTA</i> <i>variant</i>	<i>Transport</i> <i>Activity</i>	<i>Intracellular</i> <i>pH</i>	<i>LC3B</i> <i>Accumulation</i>	<i>Autophagy</i> <i>Flux</i>	<i>Cellular</i> <i>Lysosome</i> <i>size</i>	<i>Cellular</i> <i>lysosome</i> <i>number</i>	<i>Cellular</i> <i>mitochondrial</i> <i>abundance</i>	<i>Rescued</i> <i>Autophagy</i>
R295H	Unchanged	Unchanged	Unchanged	N/A	N/A	N/A	N/A	N/A
S525F	Reduced	Increased	Increased	Blocked downstream	Unchanged	Increased	Increased	Yes
R589H	Slightly Reduced	Increased	Increased	Blocked downstream	Increased	Unchanged	Increased	Yes

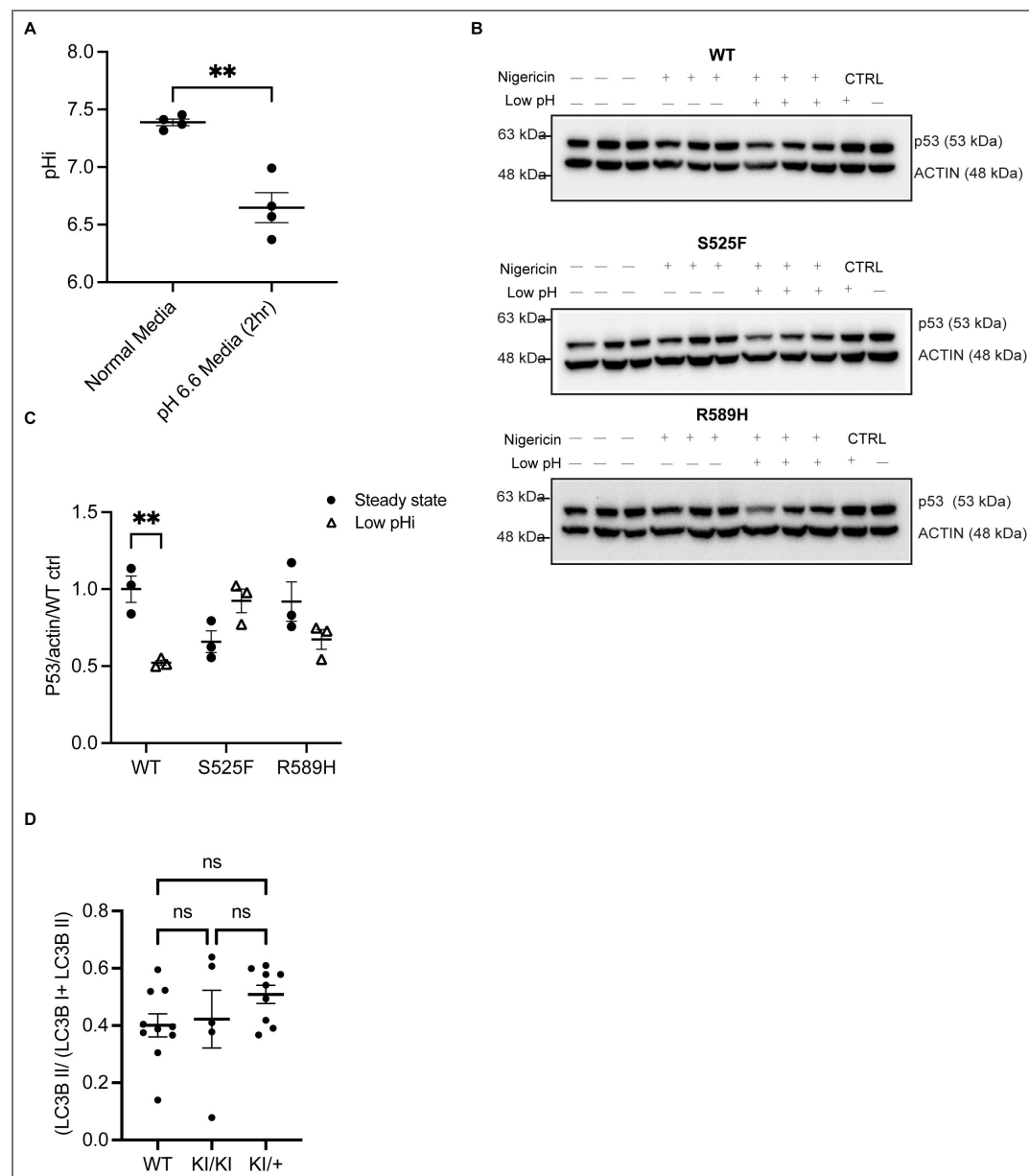
N/A, not applicable.

Supplementary Table 1. Summary of findings in dRTA kAE1 variant expressing cells compared to WT.

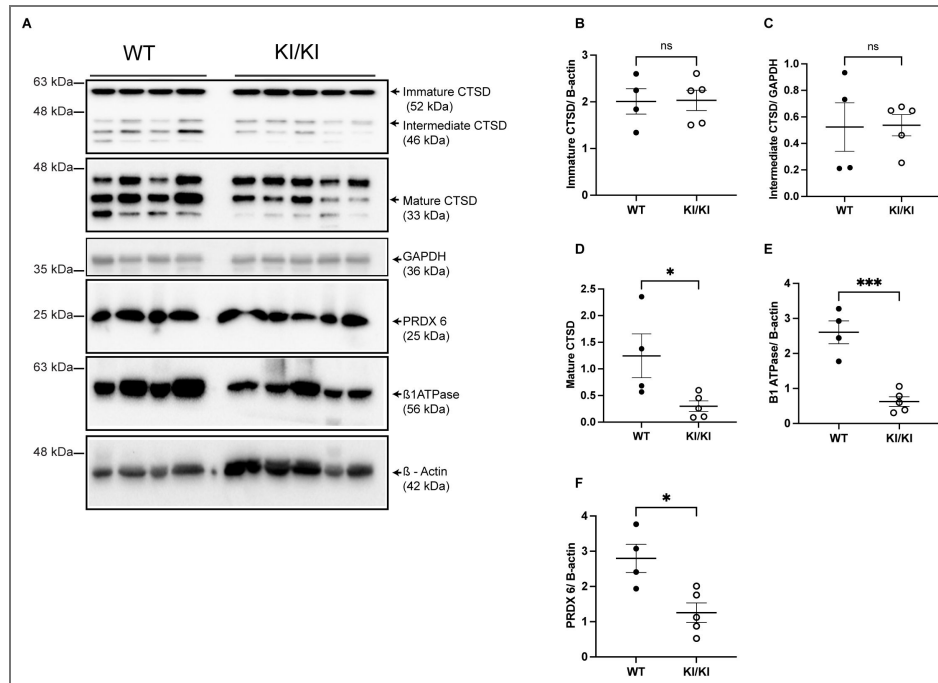
<i>Genotype</i>	<i>LC3B Accumulation</i>	<i>Intercalated Cell Lysosome size</i>	<i>Intercalated Cell lysosome number</i>	<i>Intercalated Cell mitochondrial abundance</i>
<i>R607H KI/KI</i>	Increased total LC3B	Increased	Increased	Increased

Supplementary Table 2. Summary of findings in intercalated cells from dRTA R607H knock-in (KI) relative to WT mice.

Supplementary Figures

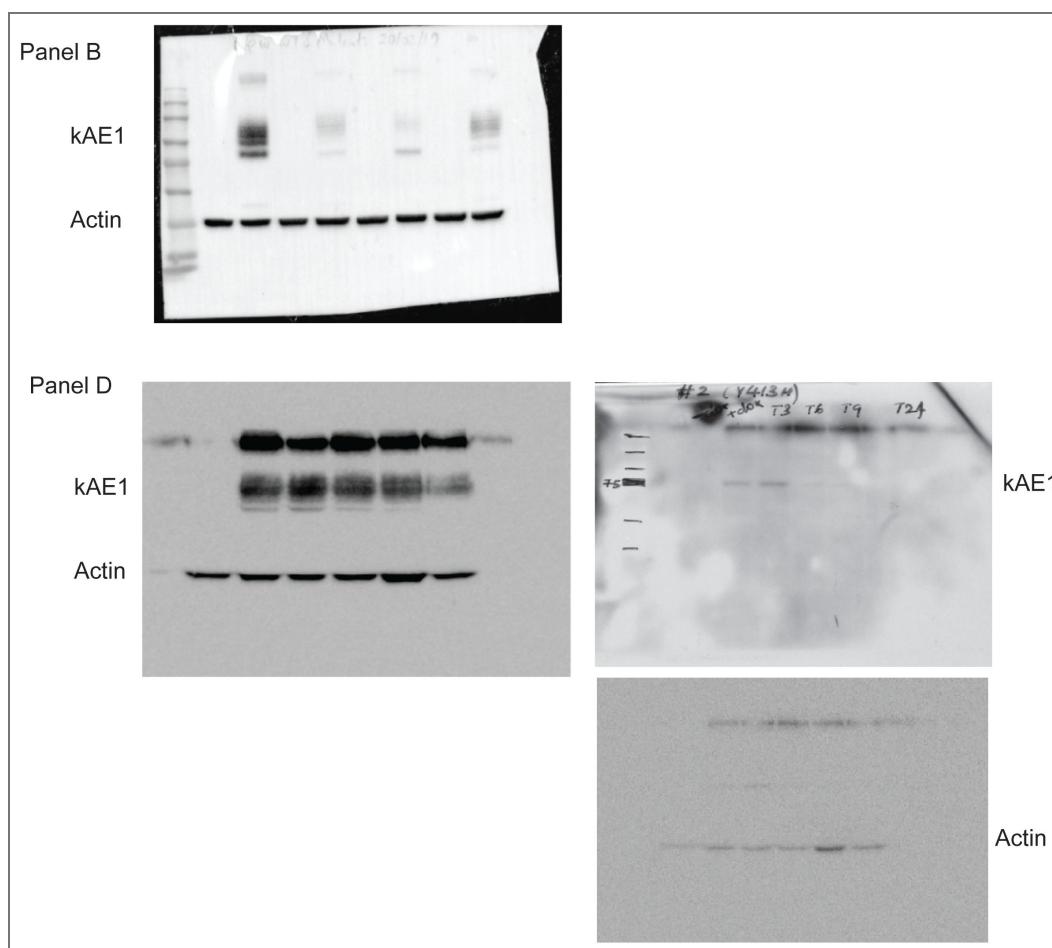


Supplementary Figure 1. Two-hour incubation in pH 6.6 media with 0.033 μ M nigericin reduces cytosolic pH to 6.9. (A) pHi of mIMCD3 cells incubated in either normal pH media or pH 6.6 media with 0.033 μ M nigericin. Error bars correspond to mean $\pm$ SEM, n= minimum 4. ** P<0.01, using Student's t test. (B) Immunoblots of p53 with corresponding actin abundance in mIMCD3 cells at steady state and under chemically reduced intracellular pH conditions. (C) Graphical representation of ratio of p53 at steady state versus at low pHi in mIMCD3 KAE1 WT, S525F and R589H cells. Black circles indicate steady state cells and triangles indicate low pHi cells. Error bars correspond to mean $\pm$ SEM, n= 3. ** indicate P<0.01 using two-way ANOVA.



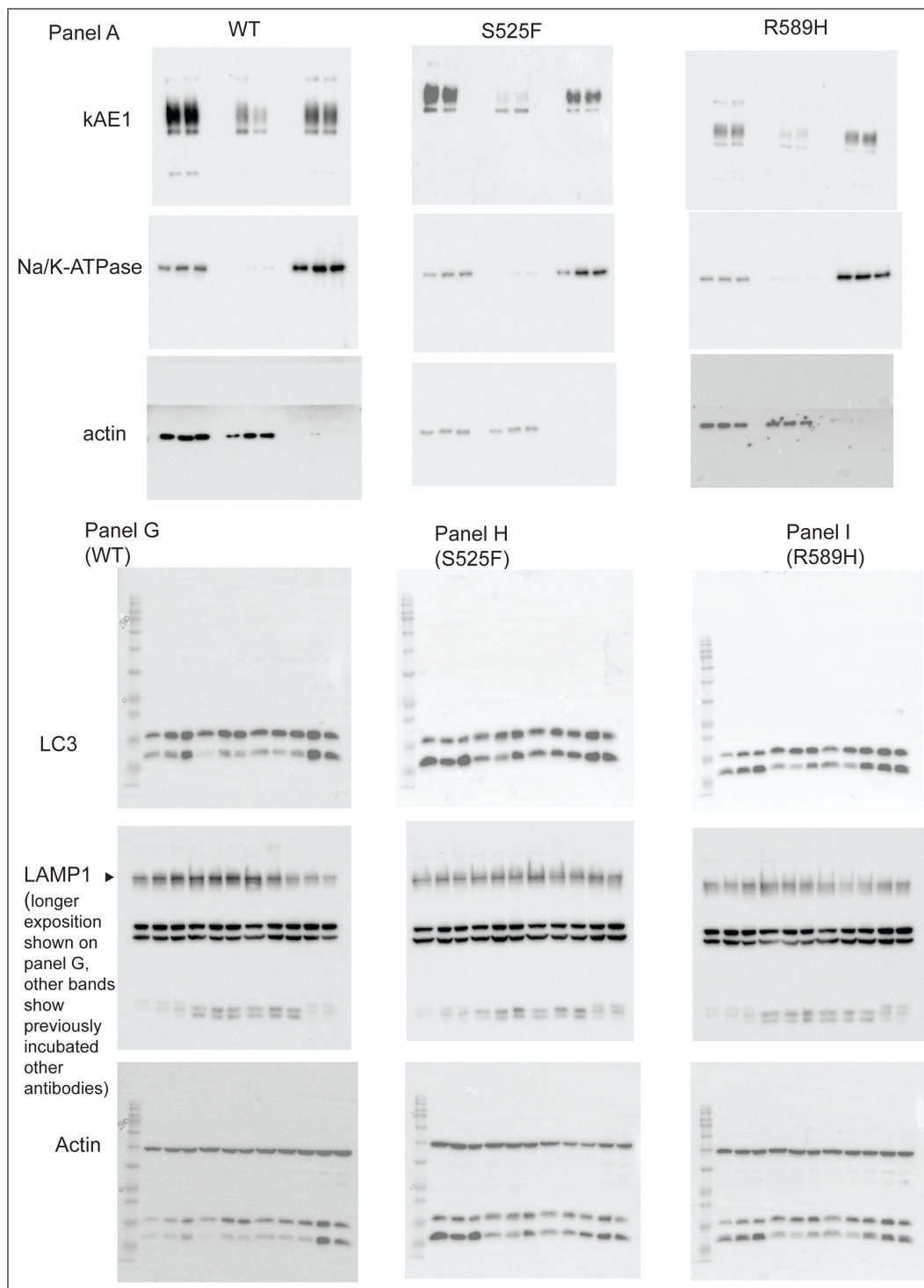
Supplementary Figure 2. dRTA kAE1 R607H knockin mice have lower abundance of intercalated cell marker B1 H⁺-ATPase, lysosomal protease Cathepsin D and antioxidant protein PRDX 6 in intercalated cells-enriched primary cells.

(A) Immunoblot of primary cells showing protein abundance of immature, intermediate and mature Cathepsin D (CTSD), antioxidant protein PRDX6 and intercalated cell marker B1 H⁺-ATPase with corresponding GAPDH and actin abundance. The first two panels are from the same blot but the second one is cropped on the region around 48 kDa and lower, and exposed longer. Primary cells were obtained by magnetic sorting and lysed upon collection. (B-F) Graphical representation of protein abundance of immature CTSD, intermediate CTSD, mature CTSD, B1 H⁺-ATPase and PRDX6 respectively in WT and knockin intercalated cell-enriched primary cells. Error bars correspond to mean ± SEM, n= minimum 4. * P<0.05, *** P<0.005 using Student's t test.



Supplementary Figure 3. Original immunoblots used for generating Figure 1 [↗](#).





Supplementary Figure 5. Original immunoblots used for generating Figure 3.

Data availability

Original blots shown in Figures 1, 2 and 3 are in the Supplementary Material file. All other relevant data supporting the findings of this study are available from the corresponding author upon reasonable request.

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Deutscher Akademischer Austauschdienst (DAAD)	PhD Scholarship	Sarder MA Hasib

Canadian Institutes of Health Research (CIHR)	PS165816	Nicolas Touret
Natural Sciences and Engineering Research Council of Canada (NSERC)	RGPIN-2018-05783	Nicolas Touret
University of Alberta (UofA)	Multiple Scholarships	Forough Chelangarimiyandoab Grace Essuman Midhat Rizvi Priyanka Mungara
Canadian Institutes of Health Research (CIHR)	Canada Graduate Scholarship-Master's	Midhat Rizvi
Natural Sciences and Engineering Research Council of Canada (NSERC)	CREATE Graduate Studentship	Shahid AKM Ullah

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Peer reviews

Reviewer #1 (Public review):

Summary:

This study is an evaluation of patient variants in the kidney isoform of AE1 linked to distal renal tubular acidosis. Drawing on observations in the mouse kidney, this study extends findings to autophagy pathways in a kidney epithelial cell line.

Strengths:

Experimental data are convincing and nicely done.

The revised manuscript incorporates most of the reviewer recommendations and presents a more cohesive story that is easier to read and assess. The data are convincing, of suitable quality and nicely presented. Statistical evaluation is rigorous. The link between kAE1 mutants and cell metabolism and autophagy is novel and provides insights on pathological observations in dRTA.

<https://doi.org/10.7554/eLife.108253.2.sa3>

Reviewer #2 (Public review):

Context and significance:

Distal renal tubular acidosis (dRTA) can be caused by mutations in a Cl-/HCO₃⁻ exchanger (kAE1) encoded by the SLC4A1 gene. The precise mechanisms underlying the pathogenesis of the disease due to these mutations is unclear, but it is thought that loss of the renal intercalated cells (ICs) that express kAE1 and/or aberrant autophagy pathway function in the remaining ICs may contribute to the disease. Understanding how mutations in SLC4A1 affect cell physiology and cells within the kidney, a major goal of this study, is an important first step to unraveling the pathophysiology of this complex heritable kidney disease.

Summary:

The authors identify a number of new mutations in the SLC4A1 gene in patients with diagnosed dRTA that they use for heterologous experiments in vitro. They also use a dRTA mouse model with a different SLC4A1 mutation for experiments in mouse kidneys. Contrary to previous work that speculated dRTA was caused mainly by trafficking defects of kAE1, the

authors observe that their new mutants (with the exception of Y413H) traffic and localize at least partly to the basolateral membrane of polarized heterologous mIMCD3 cells, an immortalized murine collecting duct cell line. They go on to show that the remaining mutants induce abnormalities in the expression of autophagy markers and increased numbers of autophagosomes, along with an alkalinized intracellular pH. They also reported that cells expressing the mutated kAE1 had increased mitochondrial content coupled with lower rates of ATP synthesis. The authors also observed a partial rescue of the effects of kAE1 variants through artificially acidifying the intracellular pH. Taken together, this suggests a mechanism for dRTA independent of impaired kAE1 trafficking and dependent on intracellular pH changes that future studies should explore.

Strengths:

The authors corroborate their findings in cell culture with a well characterized dRTA KI mouse and provide convincing quantification of their images from the in vitro and mouse experiments. The data largely support the claims as stated. Some of the mutants induce different strengths of effects on autophagy and the various assays than others, and it is not clear why this is from the data in the manuscript. The authors provide discussion of potential reasons for these differences that future studies could explore.

Weaknesses:

The pH effects of their mutants are only explored in vitro, and the in vitro system has a number of differences from a living mouse kidney or ex vivo kidney slice.

<https://doi.org/10.7554/eLife.108253.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors have identified novel dRTA causing SLC4A1 mutations and studied the resulting kAE1 proteins to determine how they cause dRTA. Based on a previous study on mice expressing the dRTA kAE1 R607H variant, the authors hypothesize that kAE1 variants cause an increase in intracellular pH which disrupts autophagic and degradative flux pathways. The authors clone these new kAE1 variants and study their transport function and subcellular localization in mIMCD cells. The authors show increased abundance of LC3B II in mIMCD cells expressing some of the kAE1 variants, as well as reduced autophagic flux using eGFP-RFP-LC3. These data, as well as the abundance of autophagosomes, serve as the key evidence that these kAE1 mutants disrupt autophagy. Furthermore, the authors demonstrate that decreasing the intracellular pH abrogates the expression of LC3B II in mIMCD cells expressing mutant SLC4A1. Lastly, the authors argue that mitochondrial function, and specifically ATP synthesis, is suppressed in mIMCD cells expressing dRTA variants and that mitochondria are less abundant in AICs from the kidney of R607H kAE1 mice. Overall, the authors provide evidence about how new kAE1 mutants may cause dRTA.

Strengths:

The authors cloned novel dRTA causing kAE1 mutants into expression vectors to study the subcellular localization and transport properties of the variants. The immunofluorescence images are generally of high quality and the authors do well to include multiple samples for all of their western blots.

<https://doi.org/10.7554/eLife.108253.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This study is an evaluation of patient variants in the kidney isoform of AE1 linked to distal renal tubular acidosis. Drawing on observations in the mouse kidney, this study extends findings to autophagy pathways in a kidney epithelial cell line.

Strengths:

Experimental data are convincing and nicely done.

Thank you

Weaknesses:

Some data are lacking or not explained clearly. Mutations are not consistently evaluated throughout the study, which makes it difficult to draw meaningful conclusions.

We have revised our manuscript to clarify some earlier explanations and provided rationale for focusing on specific variants throughout the study.

Reviewer #2 (Public review):

Context and significance:

Distal renal tubular acidosis (dRTA) can be caused by mutations in a Cl-/HCO₃- exchanger (kAE1) encoded by the SLC4A1 gene. The precise mechanisms underlying the pathogenesis of the disease due to these mutations are unclear, but it is thought that loss of the renal intercalated cells (ICs) that express kAE1 and/or aberrant autophagy pathway function in the remaining ICs may contribute to the disease. Understanding how mutations in SLC4A1 affect cell physiology and cells within the kidney, a major goal of this study, is an important first step to unraveling the pathophysiology of this complex heritable kidney disease.

Summary:

The authors identify a number of new mutations in the SLC4A1 gene in patients with diagnosed dRTA that they use for heterologous experiments in vitro. They also use a dRTA mouse model with a different SLC4A1 mutation for experiments in mouse kidneys. Contrary to previous work that speculated dRTA was caused mainly by trafficking defects of kAE1, the authors observe that their new mutants (with the exception of Y413H, which they only use in Figure 1) traffic and localize at least partly to the basolateral membrane of polarized heterologous mIMCD3 cells, an immortalized murine collecting duct cell line. They go on to show that the remaining mutants induce abnormalities in the expression of autophagy markers and increased numbers of autophagosomes, along with an alkalinized intracellular pH. They also reported that cells expressing the mutated kAE1 had increased mitochondrial content coupled with lower rates of ATP synthesis. The authors also observed a partial rescue of the effects of kAE1 variants through artificially acidifying the intracellular pH. Taken together, this suggests a mechanism for dRTA independent of impaired kAE1 trafficking and dependent on intracellular pH changes that future studies should explore.

Strengths:

The authors corroborate their findings in cell culture with a well-characterized dRTA KI mouse and provide convincing quantification of their images from the in vitro and mouse experiments

Thank you

Weaknesses:

The data largely support the claims as stated, with some minor suggestions for improving the clarity of the work. Some of the mutants induce different strengths of effects on autophagy and the various assays than others, and it is not clear why this is from the present manuscript, given that they propose pHi and the unifying mechanism

We have modified our manuscript to discuss the various strengths of the mutants and emphasize that alteration of cytosolic pH by kAE1 variants may not be the only mechanism leading to dRTA.

Reviewer #3 (Public review):

Summary:

The authors have identified novel dRTA causing SLC4A1 mutations and studied the resulting kAE1 proteins to determine how they cause dRTA. Based on a previous study on mice expressing the dRTA kAE1 R607H variant, the authors hypothesize that kAE1 variants cause an increase in intracellular pH, which disrupts autophagic and degradative flux pathways. The authors clone these new kAE1 variants and study their transport function and subcellular localization in mIMCD cells. The authors show increased abundance of LC3B II in mIMCD cells expressing some of the kAE1 variants, as well as reduced autophagic flux using eGFP-RFP-LC3. These data, as well as the abundance of autophagosomes, serve as the key evidence that these kAE1 mutants disrupt autophagy. Furthermore, the authors demonstrate that decreasing the intracellular pH abrogates the expression of LC3B II in mIMCD cells expressing mutant SLC4A1. Lastly, the authors argue that mitochondrial function, and specifically ATP synthesis, is suppressed in mIMCD cells expressing dRTA variants and that mitochondria are less abundant in AICs from the kidney of R607H kAE1 mice. While the manuscript does reveal some interesting new results about novel dRTA causing kAE1 mutations, the quality of the data to support the hypothesis that these mutations cause a reduction in autophagic flux can be improved. In particular, the precise method of how the western blots and the immunofluorescence data were quantified, with included controls, would enhance the quality of the data and offer more supportive evidence of the authors' conclusions.

Strengths:

The authors cloned novel dRTA causing kAE1 mutants into expression vectors to study the subcellular localization and transport properties of the variants. The immunofluorescence images are generally of high quality, and the authors do well to include multiple samples for all of their western blots.

Thank you

Weaknesses:

Inconsistent results are reported for some of the variants. For example, R295H causes intracellular alkalinization but also has no effect on intracellular pH when measured by

BCECF. The authors also appear to have performed these in vitro studies on mIMCD cells that were not polarized, and therefore, the localization of kAE1 to the basolateral membrane seems unlikely, based upon images included in the manuscript. Additionally, there is no in vivo work to demonstrate that these kAE1 variants alter intracellular pH, including the R607H mouse, which is available to the authors. The western blots are of varying quality, and it is often unclear which of the bands are being quantified. For example, LAMP1 is reported at 100kDa, the authors show three bands, and it is unclear which one(s) are used to quantify protein abundance. Strikingly, the authors report a nonsensical value for their quantification of LCRB II in Figure 2, where the ratio of LCRB II to total LCRB (I + II) is greater than one. The control experiments with starvation and bafilomycin are not supportive and significantly reduce enthusiasm for the authors' findings regarding autophagy. There are labeling errors between the manuscript and the figures, which suggest a lack of vigilance in the drafting process.

The R295H variant was identified in a dRTA patient and as such, it was important to report it. However, this is the first mutation located in the amino-terminus of the protein, which may be involved in protein-protein interactions, so other mechanisms may cause dRTA for this variant. We have therefore modified our manuscript to state that alteration of cytosolic pH may not be the only mechanism leading to dRTA. At this time, we are not able to measure cytosolic pH in vivo and hope to be able to do it in the future.

In our revised manuscript, we also show cell surface biotinylation results supporting that plasma membrane abundance of the kAE1 S525F and R589H variants is not significantly different than WT in non-polarized mIMCD3 cells (Figure 3 A&B), in line with the predominant basolateral localization of the variants in polarized cells (Figure 1C). Therefore, these two mutant proteins are not mis-trafficked in non-polarized cells. Finally, we have clarified which bands have been used for quantification and corrected quantifications (including ratio measurements).

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) R295H is recessively inherited, whereas Y413H is dominantly inherited: this is interesting and may be linked to their cellular expression and function. Is this information known for the other mutations examined in this study?

The S25F and R589H dRTA variants have both been reported to exhibit autosomal dominant inheritance. This information is now updated in lines 146 and 158-159.

(2) R589H expression levels are evaluated in the Western blot of Figure 1, but localization and activity are not examined in Figure 2. However, R589H is included in autophagy experiments shown in later figures. Similarly, mutant R607H is the subject of several experiments further into the manuscript, but no initial analysis is provided for this variant.

Protein abundance and localization of the R589H mutant in mIMCD3 cells have been shown in our previous publication in Supplementary Fig 5D and Supplementary Fig 2J [1]. This now indicated on lines 158-159. Our previous paper also presented a detailed study of the R607H dRTA mutant, the mouse model corresponding to the human R589H mutation. This is now indicated on lines 70, 118-119 and 180. The present study builds upon those published findings.

(3) This inconsistency is confusing, detracts from the usefulness of the study, and makes the comparative analysis of mutations incomplete. It is difficult to extrapolate from published studies in MDCK1 cells, which show different results on trafficking.

The mIMCD3 cell line, which more closely resembles the physiology of the mouse collecting duct than MDCK cells, was selected for this study and our previous one [1]. Accordingly, the results obtained are better aligned with in vivo evidence. In contrast, differences in mutant protein expression and localization observed in other cell lines, like the MDCK cells, are likely attributable to differences in their cellular origin.

(4) In Figure 2, could the authors explain why total LC3B is graphed for the data shown in mouse lysates, whereas the ratio of bands is analysed for cell lysates? Both sets of data show the two LC3B bands.

Total LC3B levels were significantly increased in the mutant compared to WT; however, no significant difference was observed in the lipidation ratio. For this reason, that graph is not shown in the main paper but has been included in the Supplementary Figure 1D.

(5) In Figure 3, representative fluorescence images should be shown for all cell lines.

We have now included representative immunofluorescence images for all cell lines in Figure 3C.

(6) pH effects: Suggest that steady state pHi (Figure 3E) and rate of alkalization (Figure 1F) would be more effective together in Figure 1. The authors should show data for the effect of nigericin on cytoplasmic pH in Figure 3. If the rate of alkalization in the mutant cells is reduced, shouldn't the intracellular steady state pH be more acidic? A cartoon depicting the transporter activity in the cell and the expected changes in pHi would be helpful. Is there a way to activate/inhibit NHE1 and rescue the effect of the mutant kAE1? It is unclear if the link between the mutant kAE1 and mitochondrial ATP production is a consequence of the intracellular pH or an indirect effect.

We opted to keep the effect of nigericin on pHi in Supplementary Fig1A given that Figure 3 already contains 11 panels. Also, in intercalated cells, the kAE1 protein physiologically exports 1 molecule of bicarbonate in exchange of 1 chloride ion import hence a reduced transport activity would result in a more alkaline intracellular pH. To clarify this point, we have included a diagram in Figure 1E as suggested. However, to calculate the rate of intracellular alkalinisation, the transporter is functioning in the opposite direction, i.e. extruding chloride and importing bicarbonate (see methods protocol for transport assay). Therefore, in this assay (Figure 1G), a defective chloride/bicarbonate activity results in a reduced rate of intracellular alkalinisation rate. This is now explained on lines 169-172.

Disruption of NHE1 function would impair sodium homeostasis and as such, potentially affect the activity of other proteins associated with acid-base balance and autophagy in collecting duct cells. Therefore, any resulting effects may not be confidently attributed specifically to the mutant kAE1. With nigericin, we aimed to alter pHi while affecting the least possible other ion concentration. Due to space considerations, Figure 1 has been reorganised to include the rate of alkalinisation and pHi (panels F and G).

Reviewer #2 (Recommendations for the authors):

(1) The authors could improve the readability of this manuscript for a general audience by clarifying and summarizing the respective phenotype(s)/effect(s) of the different mutants in some kind of table in the main figures. It is hard to keep track of the different disease mutants alongside the KI mouse mutations, as the text frequently discusses multiple mutants at a time.

As requested, we added two tables (Supplementary Tables 1 & 2) in Supplementary files summarizing the data obtained in this study. We hope this will help the readership to keep track of each variant's phenotype.

(2) The subtitle of the results section of Figure 2 should be reworded to reflect that whole kidney lysates are used for the KI mice and not the other mutants.

As requested, the title in the Results section has been modified (lines 178-179).

(3) More discussion of why the different mutants cause different strengths of phenotypes should be included.

Different variants induce different degree of functional defects as seen in Figure 1F & G. The kAE1 R295H, the only amino acid substitution in the amino-terminal cytosol causing dRTA, does not affect the transporter's function or cells' pHi. Therefore, this variant may cause dRTA via a different pathway than transport-defective S525F or partially inactive R589H variants that both affect pHi. Our study does not exclude that dRTA may be caused by other defects than pHi alterations, including defective protein-protein interactions. This discussion is now included in the manuscript on lines 386-391.

Reviewer #3 (Recommendations for the authors):

In general, I found the subject matter of this manuscript interesting and of value to the scientific community. The interpretation of the data and how much it supports the conclusion that "kAE1 variants increases pHi which alters mitochondrial function and leads to reduced cellular energy levels that eventually attenuate energy-dependent autophagic pathways" is largely incomplete. There are significant concerns about the quantification of Western blot data. Additionally, including the R607H variant in the in vitro experiments would improve the interpretation and extrapolation of in vitro data to the kidney.

We apologize for the confusion with R589H and R607H variants. The R607H mutant is the murine ortholog to the human R589H dRTA variation. To clarify this, we have added this information on line 180, in addition to lines 118-119 and line 70.

Suggestions:

(1) Can an anion replacement experiment be performed in the mIMCD cells (no Cl⁻ or no HCO₃⁻) to determine that bicarbonate transport through AE1 is responsible for the reduced ATP rates in Figure 5? Inclusion of WT + dox control would be helpful to convince the reader of the effects.

Because Seahorse real-time cell metabolism ATP rates measurements require specific and patented buffers with un-specified compositions, it was not possible to modify the Cl⁻ or HCO₃⁻ content during the ATP measurement assay. All cell lines, including empty vector cells (EV) were treated with doxycycline; thus, WT + dox was already included. The empty vector cell line treated with doxycycline allowed the exclusion of specific effects of doxycycline on mitochondrial activity as a control. This is now clarified in Figure 5 legend, lines 655-656.

(2) Can the authors measure pHi in fresh kidney sections from the R607H mouse?

Unfortunately, we are not currently able to measure pHi in fresh kidney sections and although we recognize it would benefit greatly to our study, establishing a new collaboration to perform this measurement would significantly delay the publication of this work; therefore, these results will not be available for the present manuscript.

(3) Does pH 7.0 media have any effect on autophagy, as shown in Figure 3? Why was pH 6.6 selected?

The idea was to artificially acidify pHi in mutant cell lines (that have a steady state alkaline pHi) and assess whether this acidification corrects autophagy defects. We first determined

that incubation in cell culture medium at pH 6.6 with 0.033 μ M nigericin (final potassium concentration: 168 mM) for 2 hours provided optimal conditions, i.e. ensuring cell viability over the 2-hour period while effectively lowering intracellular pH to 6.9, as demonstrated in Supplementary Figure 1A-C.

(4) In vitro experiments should be performed on polarized cells with kAE1 properly inserted in the basolateral membrane. Experiments on subconfluent, non-polarized cells do not support the hypothesis that transport functions of AE1 initiate the cascade of events attributed to these SLC4A1 mutations.

To address this point, we have performed cell surface biotinylations on 70-80 % confluent mIMCD3 cells expressing kAE1 WT, S525F or R589H mutants and show that cell surface abundance of the mutants is not significantly different from the WT protein. This is now shown in Figure 3 A&B. As cell surface biotinylation provides a more quantitative assessment of protein cell surface abundance, we have removed the immunofluorescence images from non-polarised cells and replaced them with representative immunoblots from a cell surface biotinylation assay.

Concerns:

(1) No information about the B1 ATPase antibody used.

Now provided in Supplementary Material, ATP6V1B1 Antibody from Bicell cat#20901.

(2) No actin band in Figure 1E (as prepared).

Actin bands are provided for each blot in Figure 1D.

(3) Figures 1E and 1F are labelled wrong in the figure versus the results section.

Thank you for letting us know, this is now corrected.

(4) The cortical sections shown in Figure 4 for the KI/KI do not appear to have the morphology of a CCD. The authors may want to consider including glomeruli to convince the reader of the localization of the tubules. Same concern with Figure 5G and I. The WT image in 5G does not have the morphology of a CCD. Principal cells should be predominant, and ICs should be dispersed.

Both figures 4 and 5 have been updated with images showing glomeruli (light blue “G” on figure) with neighbour and dispersed IC staining.

(5) The quantification of LAMP1 in Figure 4 is unclear. How did the authors determine the boundary of AICs, and how did they calculate the volume of lysosomes? If a zstack was used, how are the authors sure that their 10um section includes the entire AIC?

The quantification of LAMP1 is detailed under “Image analysis”, then “Volocity” sections in Supplementary Material. The boundary of A-IC was manually detected in Volocity based on the presence of the H^+ -ATPase before Volocity analysis for lysosomal volume as described in the Methods.

The 10 micron sections are expected to include full AIC as well as partial AIC, but the frequency of these events should be the same between WT and variants’ sections, therefore they were all included in the analysis if cells displayed H^+ -ATPase signal.

(6) Figure 5: There is no description of how ATP rates are calculated from the provided traces.

We used Agilent Seahorse XF ATP rate assay kit for this experiment. In this assay, the total ATP rate is the sum of ATP production rate from both glycolysis and oxidative

phosphorylation. Glycolysis releases protons in a 1:1 ratio with ATP hence the glycolytic ATP rate is calculated from the glycolytic proton efflux rate (glycoPER). GlycoPER is determined by subtracting respiration linked proton efflux from total proton efflux by inhibiting complex I and III. This information is now added to Supplementary Material, in the “Metabolic Flux analysis” section.

(7) Figure labels in Figure 5 are wrong. It seems 5H (as presented) should actually be labeled 5G. In 5H (G?), why did some cells not have any TOM20 pixel intensity for S525F and R589H variants?

Confocal image acquisition in this experiment was kept under the same settings to allow comparison between samples. Therefore, some cells show dimer fluorescence than others. From the figure 5 panels, all cells showed TOM 20 pixel intensity. Figure 5H panel has been relabelled Figure 5G.

(8) In Figure 2, the summary graphs show analysis of more samples than are visible on the included western blots. What is the rationale for this? Why does S525F have 9 samples in BafA1 while R295H only has 3 (2H)? Yet, R295H has 6 samples in 2I. In 2D, S525F has at least 9 samples. Explain.

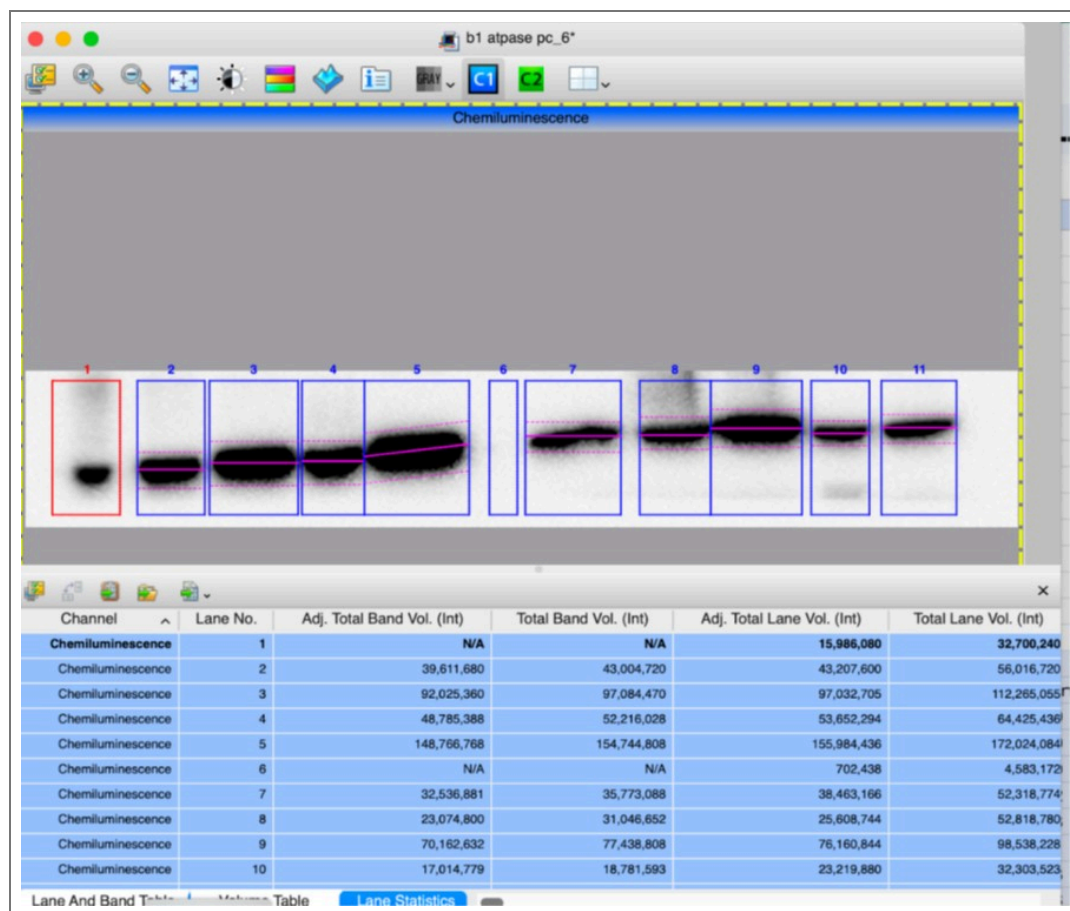
Figure 2A-C shows representative immunoblots, among several ones independently conducted. Therefore, the final number of samples is higher than showed on Figure 2. This is now indicated in Figure 2 legend, line 603. It became clear quite early in our study that the recessive kAE1 R295H variant does not behave similarly to the other variants studied, maybe because it affects the cytosolic domain, so we did not perform as many replicates for this variant as we did for the others. However, we felt it was valuable to the research community to report the characterization of this variant and decided to keep it in our study.

(9) In general, the actin loading does not appear to be equal between samples. And some figures show the same actin blot twice (2A, C) while some show independent actin bands for LC3B and p62. Equal loading seems a fairly significant control, considering the importance of quantification in the figures.

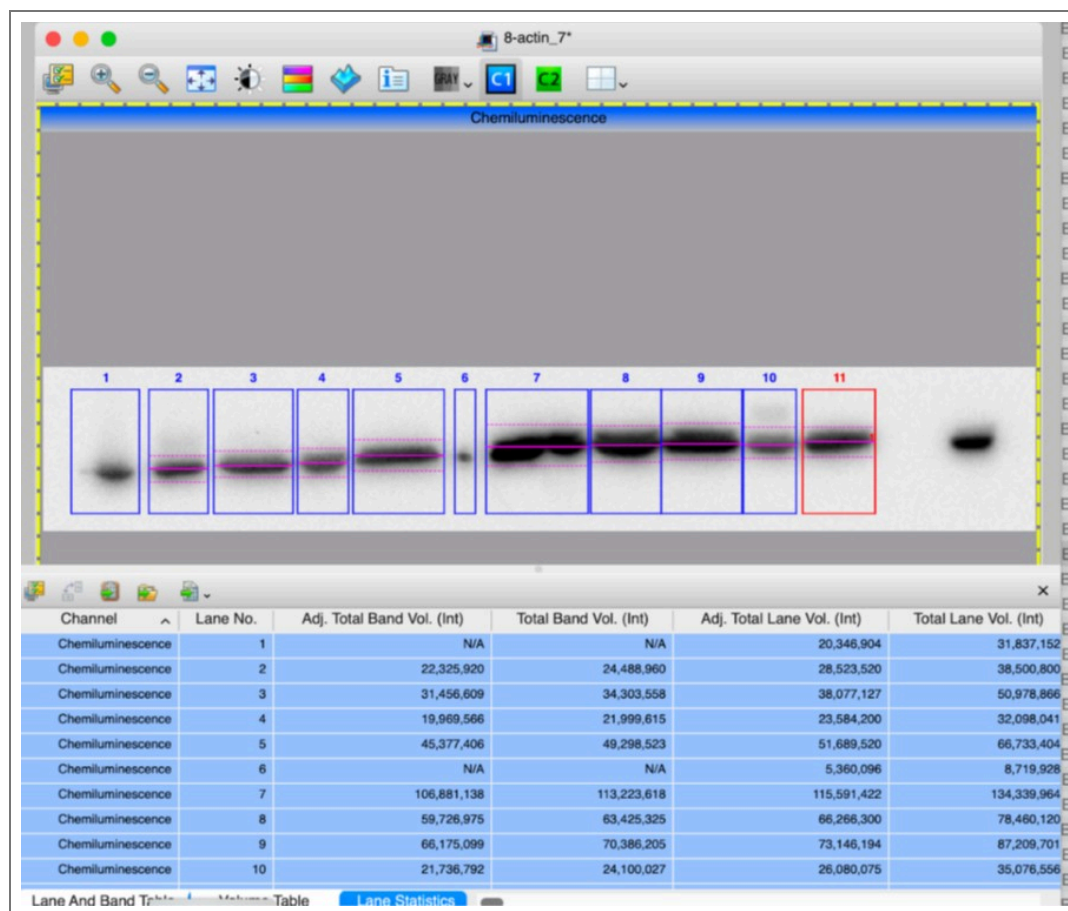
In addition to performing protein assays, we systematically conduct immunoblot with anti-b-actin antibody to control for loading variability. When possible, two or three proteins, including actin, are detected on the same blot, when molecular weight differ enough. This sometimes results in b-actin being used as a loading control for two different proteins, as seen on Figure 2A and 2C. This is now indicated on lines 605606.

(10) In the Supplemental Figure 2, which band is being quantified for mature CTSD at 33kDa? Same for intermediate CTSD. The quantification of V-ATPase seems questionable based on the actin variance shown in the blot. Surely the ratio of the fourth sample is greater than 1.

Supplementary Figure 2 has been updated to include arrows indicating which band was selected for the quantification. After verifying the measurements of band intensities from “Image Lab” quantification software, we confirm the results, including that fourth KI/KI sample has a ratio of 0.78 (Adj Total Band Vol (Int), lanes 10). Screen shots of quantifications are attached below.



Author response image 1.



Author response image 2.

(11) Why are the experiments performed on non-confluent IMCD cells? Figure 1D shows good basolateral localization of AE1, yet the other experiments in the manuscript appear to use IMCD cells in low confluent states, without proper localization of AE1. Figure 3A shows AE1 dispersed throughout the cytoplasm. Why have the authors decided to study the effects of an anion exchanger without it being properly localized to the basolateral membrane? Shouldn't all experiments be performed in polarized IMCDs? If AE1 isn't properly in the membrane, and the cells do not have defined apico-basolateral polarity, then what role can AE1-mediated intracellular pH change have on the results of the experiments? Were the pH experiments in 3E performed on polarized cells? Or even 1F?

To address this point, we have performed cell surface biotinylations on 70-80 % confluent mIMCD3 cells expressing kAE1 WT, S525F or R589H mutants and show that cell surface abundance of the mutants is not significantly different from the WT protein. This is now shown in Figure 3A & B. As it provides a more quantitative assessment of protein cell surface abundance, we have removed the immunofluorescence images from non-polarised cells and replaced them with a representative immunoblot from a cell surface biotinylation assay.

(12) As mentioned in the public comments, how is the ratio $A/(A+B)$ greater than 1? With A and $B > 0$. In Figure 3, the data is reasonable, but in Figure 2, the data is simply impossible. What is the explanation for this phenomenon? Why was this presentation of data approved? Is it supposedly a fold of WT, like 2K and 2L? Is the reader also to believe that total LC3B is 2-fold greater in KI/KI mice, as shown in 2K? My eyes, though not

densitometry equipment, cannot confirm this. The actin bands are not equal. Yet again, there are 4 lanes of KI/KI mice, but the quantification shows 5 samples.

The ratios in figure 2D, 2F, 2H and 2L have been re-calculated and corrected. As indicated above, immunoblots are representative and quantification of additional blots has been included in the graphs.

(12) Spelling error Figure 4B: cels.

Corrected

References

(1) Mumtaz, R. et al. Intercalated Cell Depletion and Vacuolar H⁺-ATPase Mistargeting in an Ae1 R607H Knockin Model. *Journal of the American Society of Nephrology* 28, 1507–1520 (2017).

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